Lecture 5, Repeated Games: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 25 exercises of `Listas/pratica-lecture-05.md`

November 2026

These are detailed solutions to the 25 exercises of `Listas/pratica-lecture-05.md` (Lecture 5, *Repeated games*), in the list's order: three tiers (Foundations, Exam level, Advanced) with the difficulty rising from 1 (apply a definition) to 5 (graduate level). Each heading shows the item's number in the list, its level and its difficulty. Each statement is printed verbatim from the list right before its solution. Each solution has the same parts: (a) what is asked; (b) the method, with the book section and page and the Lecture 5 slide; (c) the solution, every step shown; (d) the economic interpretation; (e) the verification, by `check_L05.py` in this folder (new solutions) or `check_C_L05.py` in `lista-reais-lectures-01-05-solutions/` (the eight reused ones: items 1, 5, 6, 10, 15, 16, 17, 21), and against the book's worked answer where one exists; (f) common mistakes. Method of the slides: payoffs of infinitely repeated games are normalised, $U_i = (1-\delta)\sum_t \delta^{t-1} u_i^t$; the one-stage deviation principle is checked history by history, on and off the path; thresholds follow $\delta^* = (v^d - v^c)/(v^d - v^*)$ (Tadelis (10.5)). In the finitely repeated games (two stages) payoffs are $u_i^1 + \delta u_i^2$. Three items deserve a warning: item 8 (Tadelis 9.3) has no solution for $\delta \leq 1$ as printed, item 9 (Cabral 7.12) has no tension as printed, and item 20 (MWG 12.AA.1) is false as printed (the MWG manual's erratum); each solution says what is true and why. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG), and Cabral (2017). In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies. The eight reused solutions were written before the slides existed and cite "Lecture 4, slides 21–22" for the repeated-game set-up; the matching slides of the real lecture are Lecture 5, slides 17–26.

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Lecture 5 — Repeated games

Item 1: Cabral Exercise 7.13

Foundations · difficulty 1/5

Statement (verbatim from the list, item 1)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 7 Exercises, Exercise 7.13 "Finitely repeated game", p. 182 [PDF 194]. Answer in the source: none in the book. Reproduced verbatim; Figure 7.1 (p. 160 [PDF 172]), which the exercise refers to, is added below it.

> **7.13. Finitely repeated game.** Suppose that the game depicted in Figure 7.1 is repeated T times, where T is known. Show that the only subgame perfect equilibrium is for players to choose (B, R) in every period.

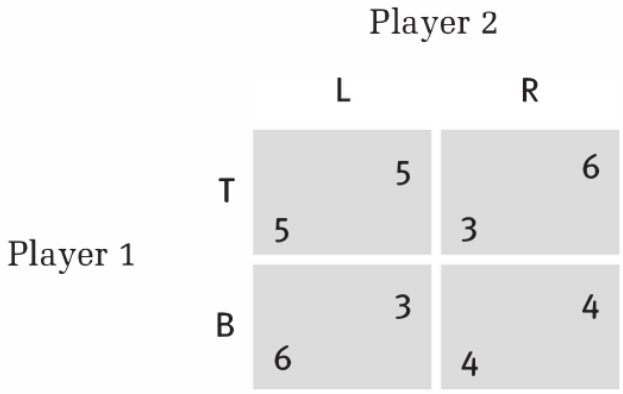


FIGURE 7.1
The "prisoner's dilemma" game

Figure 7.1 The "prisoner's dilemma" game (p. 160). In each cell, Player 1's payoff is in the lower left corner and Player 2's in the upper right corner.

(a) What is asked. Prove that when Cabral's Prisoner's Dilemma (Figure 7.1) is repeated a known number T of times, the only subgame-perfect equilibrium (SPE) has (B, R) played in every period.

(b) Method. Backward induction on the finitely repeated game $G(T, \delta)$ (Lecture 4, slide 22: each period starts a new subgame; slide 17: an SPE is a Nash equilibrium in every subgame; slides 9–10: backward induction). The result is Tadelis Prop. 10.1 (§10.1, p. 192), a special case of Prop. 9.3 (§9.4, p. 181): if the stage game has a unique Nash equilibrium, the finitely repeated game has a unique SPE, the stage equilibrium in every period after every history. Cabral sets up the repeated Prisoner's Dilemma in §7.3 (pp. 172–174). The proof is an induction on the number of periods left.

(c) Solution. Notation. Periods $t = 1, \ldots, T$; h^{t-1} is the history of action profiles before period t; player i's payoff is $\sum_{t=1}^{T} \delta^{t-1} u_i(a^t)$ with $\delta \in (0, 1]$ (slide 22). Stage payoffs (Figure 7.1, player 1 first):

	L	R
T	5, 5	3, $\underline{6}$
B	$\underline{6}$, 3	$\underline{4}$, $\underline{4}$

Step 1: the stage game. For player 1, B beats T against both columns: $6 > 5$ against L and $4 > 3$ against R. So B is strictly dominant; by symmetry R is strictly dominant for player 2. The stage game therefore has exactly one Nash equilibrium, (B, R) with payoffs $(4, 4)$, and no mixed equilibrium (a strictly dominated action gets zero weight in any equilibrium).

Step 2: the last period, $t = T$. Take any history h^{T-1}. The payoffs already earned, $\sum_{t<T} \delta^{t-1} u_i(a^t)$, are fixed and nothing follows period T. So the subgame after h^{T-1} is the stage game with every payoff multiplied by $\delta^{T-1} > 0$ and a constant added. Neither operation changes best replies, so the only Nash equilibrium of that subgame is (B, R). An SPE must therefore play (B, R) in period T after *every* history.

Step 3: the induction step. Suppose that in every SPE, (B, R) is played in periods $k+1, \ldots, T$ after every history. Take any history h^{k-1}. Whatever is played in period k, the continuation from $k+1$ on is (B, R) in each of the $T - k$ remaining periods, worth

$$W = \sum_{t=k+1}^{T} \delta^{t-1} \cdot 4,$$

the same constant for both players and independent of the period-k action. Period k's subgame therefore reduces to the stage game scaled by δ^{k-1} plus the constant W (plus the sunk past payoffs). Its unique Nash equilibrium is again (B, R). The claim holds for period k after every history.

Step 4: conclusion. By induction from $k = T$ down to $k = 1$, the unique SPE is: each player plays B (player 1) and R (player 2) in every period after every history. Its payoff is $4 \sum_{t=1}^{T} \delta^{t-1}$ for each player. The argument uses neither the value of δ nor the value of T, only that T is finite and known.

One-stage deviation check. At any history in period k, a single deviation to T (or L) changes only the period-k payoff, from 4 to 3 if the opponent plays R, while the continuation stays (B, R). The deviation loses δ^{k-1}, so no one-stage deviation pays (Tadelis Thm. 9.1, p. 185).

A stronger statement (Nash equilibrium paths). Even without subgame perfection, every Nash equilibrium of the repeated game has the *path* (B, R) in every period. Player 1 can always play B, which gives him at least 4 per period (row B pays 6 or 4), and 4 is also what (B, R) pays. Going backwards along the path: if the path had T (or L) in some period, the player could switch to B (or R) there and in every later period, earn one more in that period, and get at least 4 later, which is what the path gives later by the induction. The punishment threats that could deter this are worth nothing, because the minimax payoff, 4, equals the equilibrium payoff. `check_C_L05.py` confirms this for $T = 2$ by brute force.

(d) Interpretation. Cooperation in a finitely repeated game needs a threat for the future, and here there is none. In the last period there is no future, so the players defect. In the second-to-last period the last period's outcome is already fixed, so there is again nothing to protect, and the argument unravels back to period 1. Cabral's point in §7.3 is that an *indefinitely* repeated game, one that ends after each period with probability $1 - \alpha$ (Cabral's notation, p. 172) and so has no known last period, is needed for retaliation to sustain (T, L). A known end date, however distant, removes it.

(e) Verification. `check_C_L05.py` checks that B and R are strictly dominant and that (B, R) is the unique Nash equilibrium of the stage game and of the folded game with any added constant.

For $T = 2$ it enumerates all $32 \times 32 = 1024$ pure strategy profiles (a period-1 action and a map from the four period-1 outcomes to a period-2 action) at $\delta \in \{\frac{1}{10}, \frac{1}{2}, 1\}$. It finds exactly one SPE (always B, always R), and every Nash equilibrium has the path $(B, R), (B, R)$. Output: "Cabral 7.13: unique stage NE (B,R); T=2 brute force (1024 profiles x 3 deltas): 1 SPE, all NE paths (B,R),(B,R)". Cabral prints no answer; the result matches Tadelis Prop. 10.1.

(f) Common mistakes.

- Showing only the outcome, "(B, R) every period", and not the strategy "(B, R) after *every* history". An SPE prescribes play off the path too, and the induction must cover every history.
- Arguing that δ large enough or T long enough allows cooperation. With a known last period and a unique stage equilibrium, neither matters.
- Folding the game forward ("if they cooperate today they will cooperate tomorrow") instead of starting at period T. Backward induction runs from the end.

Answer. B and R are strictly dominant, so (B, R) is the unique stage equilibrium. In period T every subgame is the stage game, so (B, R) is played after every history; given that, the continuation from any period k on does not depend on period-k actions, so period k is again the stage game plus a constant and (B, R) is played. By induction the unique SPE plays (B, R) in every period after every history, for any T and any δ.

Item 2: Cabral Exercise 7.8

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 7 Exercises, Exercise 7.8 "Finitely repeated game", p. 180 [PDF 192]. Answer in the source: none in the book. Reproduced verbatim.

> **7.8. Finitely repeated game.** Consider a one-shot game with two equilibria and suppose this game is repeated twice. Explain in words why there may be equilibria in the two-period game which are different from the equilibria of the one-shot game.

(a) What is asked. Explain in words why a one-shot game with two equilibria, played twice, can have equilibria whose play differs from the repetition of the one-shot equilibria.

(b) Method. Finitely repeated games with several stage equilibria (Tadelis §10.1, pp. 190–192, and the one-stage deviation principle, Thm. 9.1, p. 185; Gibbons §2.3.A, pp. 82–88). Slides, Lecture 5: stage-game equilibria and the repeated game (slide 10), the $T = 2$ example with two equilibria (slides 11–13), one-stage deviation principle (slide 14), discounting (slide 15).

(c) Solution. Step 1: the last period. The subgame after any first-period outcome is the stage game itself, so every SPE plays a stage-game Nash equilibrium there (slide 10). With two equilibria, e^{good} and e^{bad}, the players can choose *which one* after each first-period outcome.

Step 2: the first period. A first-period action profile a that is not a stage equilibrium tempts some player to deviate and gain g_i. The second period can make that deviation costly: play e^{good} after a (the carrot) and e^{bad} after a deviation (the stick). This is credible, because both are Nash equilibria of the second-period subgame. Player i then conforms if the continuation gap is larger than the temptation:

$$\delta \left[u_i(e^{\text{good}}) - u_i(e^{\text{bad}}) \right] \;\geq\; g_i .$$

When it holds, a is played in period 1 although it is not an equilibrium of the one-shot game. The repetition of one-shot equilibria (every period a stage equilibrium) is still an SPE, but it is no longer the only one.

Step 3: an example (slide 11). The stage game

	m	f	r
M	$4,\ 4$	$-1,\ 5$	$0,\ 0$
F	$5,\ -1$	$\underline{1},\ \underline{1}$	$0,\ 0$
R	$0,\ 0$	$0,\ 0$	$\underline{3},\ \underline{3}$

has two pure equilibria, (F, f) with $(1, 1)$ and (R, r) with $(3, 3)$. Consider: play (M, m) in period 1; in period 2 play (R, r) after (M, m) and (F, f) after any other outcome. The best deviation of player 1 is F (payoff 5). Conforming gives $4 + 3\delta$, deviating gives $5 + \delta$, so he conforms iff $\delta \geq \frac{1}{2}$; player 2 is symmetric. At $\delta = 1$ the comparison is $7 \geq 6$ (slide 13). So (M, m) is played in period 1 of an SPE, which no repetition of (F, f) and (R, r) produces.

Step 4: why two equilibria are needed. If the stage game has a unique equilibrium, the second period's play cannot depend on the first, the carrot and the stick coincide, and the condition in Step 2 fails: unraveling (slide 9, and Cabral 7.13, item 1).

(d) Interpretation. Repetition adds new equilibria only through the *choice among* stage equilibria: a future that depends on the present is a credible threat exactly when the threatened action is itself a best reply in the last period.

(e) Verification. `check_L05.py` computes the stage equilibria of the slide-11 game, (F, f) and (R, r), and confirms by the one-stage deviation test that (M, m) is a first-period SPE outcome for $\delta = \frac{1}{2}$, $\frac{3}{4}$, 1 and not for $\delta = 0.49$.

Answer. With two stage equilibria, period-2 play can be made to depend on period-1 play (reward with the good equilibrium, punish with the bad one) and both are best replies in the last period, so the threat is credible. A non-equilibrium first-period profile is then sustained whenever $\delta\,[u_i(e^{\text{good}}) - u_i(e^{\text{bad}})] \geq g_i$ (in the slide-11 game, $\delta \geq \frac{1}{2}$ for (M, m)). With a unique stage equilibrium nothing new arises.

Item 3: Muñoz-Garcia Ch. 4 Exercise 18

Foundations · difficulty 1/5

Statement (verbatim from the list, item 3)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 18 "Backward Induction in Wage Negotiations", pp. 200–201 [PDF 212–213]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider the following bargaining game between a labor union and a firm. In the first stage, the labor union chooses the wage level, w, that all workers will receive. In the second stage, the firm responds to the wage w by choosing the number of workers it hires, $h \in [0, 1]$. The labor union's payoff function is $u_L(w, h) = w \cdot h$, thus being increasing in wages and in the number of workers hired. The firm's profit is
>
> $$\pi(w, h) = \left(h - \frac{h^2}{2}\right) - wh$$
>
> Intuitively, revenue is increasing in the number of workers hired, h, but at a decreasing rate (i.e., $h - \frac{h^2}{2}$ is concave in h), reaching a maximum when the firm hires all workers, i.e., $h = 1$ where revenue becomes $1/2$.
>
> **Part (a)** Applying backward induction, analyze the optimal strategy of the last mover (firm). Find the firm's best response function $h(w)$.
>
> **Part (b)** Let us now move on to the first mover in the game (labor union). Anticipating the best response function of the firm that you found in part (a), $h(w)$, determine the labor union's optimal wage, w^*.

(a) What is asked. A union sets a wage w, then a firm picks the number of workers $h \in [0, 1]$. Find the firm's best response $h(w)$ and the union's optimal wage.

(b) Method. Backward induction in a two-stage game (Gibbons §2.2.A, pp. 71–73; Tadelis §9.2–9.3): solve the last mover's problem for every history, substitute it into the first mover's payoff, then maximise. Lecture 5 slide 14 (solve the last stage first, check one deviation at a time) applies the same logic to repeated games.

(c) Solution. Step 1: the firm (stage 2). For a given w the firm maximises $\pi(w, h) = h - \frac{h^2}{2} - wh$ over $h \in [0, 1]$. The first-order condition is $1 - h - w = 0$ and $\partial^2\pi/\partial h^2 = -1 < 0$, so π is concave and

$$h(w) = 1 - w \quad \text{for } w \in [0, 1].$$

For $w \geq 1$ the unconstrained solution is ≤ 0, so $h(w) = 0$. The best response is $h(w) = \max\{0,\, 1 - w\}$. The firm hires everybody at $w = 0$ and nobody at $w = 1$.

Step 2: the union (stage 1). Anticipating $h(w)$, the union's payoff is

$$u_L\big(w, h(w)\big) = w(1 - w) = w - w^2, \qquad w \in [0, 1].$$

The first-order condition $1 - 2w = 0$ gives $w^* = \frac{1}{2}$, and the second derivative is $-2 < 0$. Then $h^* = h(\frac{1}{2}) = \frac{1}{2}$.

Step 3: payoffs. $u_L = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ and $\pi = (\frac{1}{2} - \frac{1}{8}) - \frac{1}{4} = \frac{1}{8}$.

The SPE is: union plays $w^* = \frac{1}{2}$; firm plays $h(w) = \max\{0, 1 - w\}$ after every w, in particular $h(\frac{1}{2}) = \frac{1}{2}$. The strategy of the firm after $w \neq \frac{1}{2}$ matters: it is what makes the union's choice optimal.

(d) Interpretation. The union uses its first-mover advantage but cannot ignore the firm's reaction: a higher wage lowers employment. The outcome is not efficient. Joint surplus $h - h^2/2$ is largest at $h = 1$ (value $\frac{1}{2}$), whereas here it is $\frac{3}{8}$. Both sides could gain by moving to $h = 1$ with a wage in between, but nothing in a single play of the game enforces it. This is the starting point of item 24 (Gibbons 2.16), where repetition supports such outcomes.

(e) Verification. `check_L05.py` solves both stages with sympy and reproduces the book's answer: $h(w) = 1 - w$, $w^* = \frac{1}{2}$, $h^* = \frac{1}{2}$, $u_L = \frac{1}{4}$, $\pi = \frac{1}{8}$. It also confirms by grid that $w(1 - w)$ has maximum $\frac{1}{4}$.

Answer. $h(w) = 1 - w$ (and 0 for $w \geq 1$); $w^* = \frac{1}{2}$, $h^* = \frac{1}{2}$; payoffs $u_L = \frac{1}{4}$ for the union and $\pi = \frac{1}{8}$ for the firm, as in the book.

Item 4: Gibbons Problem 2.22

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems (Section 2.4), Problem 2.22, p. 138 [PDF 145]. Answer in the source: none in the book. Reproduced verbatim.

> 2.22. Give the extensive- and normal-form representations of the bank-runs game discussed in Section 2.2.B. What are the pure-strategy subgame-perfect Nash equilibria?

Referred to in the statement. The bank-runs game of Section 2.2.B (Gibbons pp. 73–74 [PDF p. 80–81]) (reproduced from the book):

> Two investors have each deposited D with a bank. The bank has invested these deposits in a long-term project. If the bank is forced to liquidate its investment before the project matures, a total of $2r$ can be recovered, where $D > r > D/2$. If the bank allows the investment to reach maturity, however, the project will pay out a total of $2R$, where $R > D$.
>
> There are two dates at which the investors can make withdrawals from the bank: date 1 is before the bank's investment matures; date 2 is after. For simplicity, assume that there is no discounting. If both investors make withdrawals at date 1 then each receives r and the game ends. If only one investor makes a withdrawal at date 1 then that investor receives D, the other receives $2r - D$, and the game ends. Finally, if neither investor makes a withdrawal at date 1 then the project matures and the investors make withdrawal decisions at date 2. If both investors make withdrawals at date 2 then each receives R and the game ends. If only one investor makes a withdrawal at date 2 then that investor receives $2R - D$, the other receives D, and the game ends. Finally, if neither investor makes a withdrawal at date 2 then the bank returns R to each investor and the game ends.
>
> In Section 2.4 we will discuss how to represent this game formally. For now, however, we will proceed informally. Let the payoffs to the two investors at dates 1 and 2 (as a function of their withdrawal decisions at these dates) be represented by the following pair of normal-form games. Note well that the normal-form game for date 1 is nonstandard: if both investors choose not to withdraw at date 1 then no payoff is specified; rather, the investors proceed to the normal-form game at date 2.
>
> *Date 1*

	withdraw	don't
withdraw	r, r	$D, 2r - D$
don't	$2r - D, D$	next stage

> *Date 2*

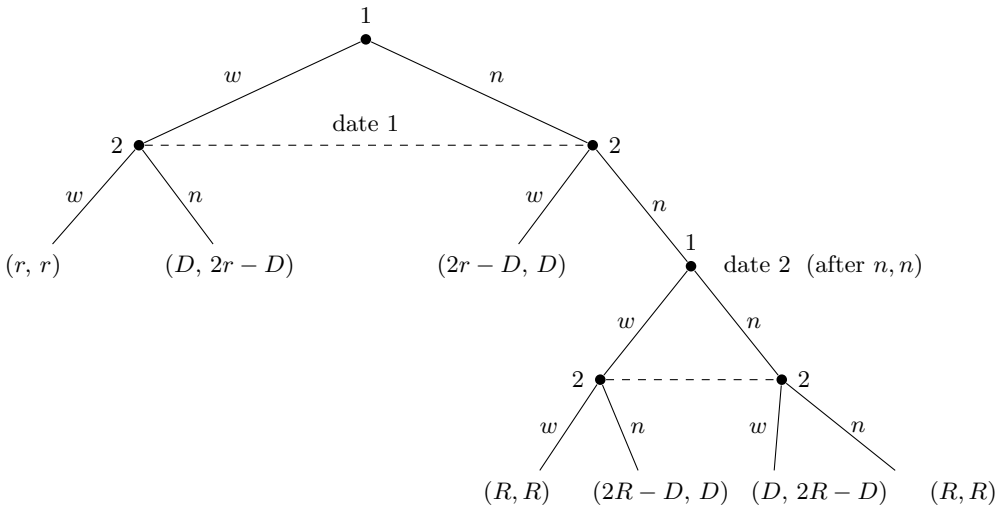
	withdraw	don't
withdraw	R, R	$2R - D, D$
don't	$D, 2R - D$	R, R

(a) What is asked. Write the bank-runs game of Section 2.2.B in extensive and normal form and find its pure-strategy SPNE.

(b) Method. Gibbons §2.2.B (pp. 73–75) and the definitions of §2.4.A–B (pp. 115–128): a strategy specifies an action at every information set, including those a player's own earlier action rules out; an SPNE is a Nash equilibrium of every subgame. Solve date 2 first (slide 14: last stage first).

(c) Solution. The game. Two investors each deposited D; liquidation at date 1 yields a total $2r$ with $D > r > D/2$, maturity yields $2R$ with $R > D$. Each investor chooses w (withdraw) or n (do not), simultaneously, at date 1 and, if the game has not ended, at date 2. There is no discounting.

Extensive form. Dashed lines join the nodes of one information set (a player does not see the other's simultaneous choice). Payoffs: investor 1 first.



Normal form. A strategy is a pair (date-1 action, date-2 action); the second letter is used only after (n, n). There are four strategies: ww, wn, nw, nn. If either investor withdraws at date 1 the date-2 letter is irrelevant, so the matrix has a block structure (underlined payoffs are best replies):

	ww	wn	nw	nn
ww	$\underline{r}, \underline{r}$	$\underline{r}, \underline{r}$	$D,\ 2r-D$	$D,\ 2r-D$
wn	$\underline{r}, \underline{r}$	$\underline{r}, \underline{r}$	$D,\ 2r-D$	$D,\ 2r-D$
nw	$2r-D,\ D$	$2r-D,\ D$	$\underline{R}, \underline{R}$	$\underline{2R-D},\ D$
nn	$2r-D,\ D$	$2r-D,\ D$	$D,\ \underline{2R-D}$	$R,\ R$

Best replies. Use $D > r > D/2$ (so $2r - D < r$) and $R > D$ (so $2R - D > R > D$). Against a rival who withdraws at date 1 (ww or wn), withdrawing gives r and waiting gives $2r - D < r$: withdraw. Against nw, withdrawing at date 1 gives $D < R$, playing nw gives R and nn gives D: the best reply is nw. Against nn, nw gives $2R - D$, the largest payoff in the column.

Nash equilibria. Reading the matrix: the four profiles in which both investors withdraw at date 1 (ww or wn for each) are equilibria with payoff (r, r), and (nw, nw) is an equilibrium with payoff (R, R). (nn, nn) is not (either investor gains by switching to nw), nor are (nw, nn) and (nn, nw). So there are five Nash equilibria.

Subgames and SPNE. The only proper subgame is the date-2 game after (n, n), where w is strictly dominant ($2R - D > R$ against w and $R > D$ against n): its unique Nash equilibrium is (w, w). An SPNE must play it, so the date-2 letter must be w. That removes every equilibrium that uses n at date 2, namely $(wn, \cdot)$ and $(\cdot, wn)$. The pure-strategy SPNE are

$$(ww,\ ww) \text{ with payoff } (r, r) \quad \text{(a run on the bank)}, \qquad (nw,\ nw) \text{ with payoff } (R, R).$$

Profiles such as (wn, wn) are Nash equilibria but not subgame perfect: they are supported by the threat of not withdrawing at date 2, which is not credible.

(d) Interpretation. This is the one-period game with two equilibria after substituting the date-2 equilibrium (R, R) for the continuation (Gibbons, Figure 2.2.1). A run is self-fulfilling: if investor 1 expects investor 2 to withdraw at date 1, withdrawing is a best reply although both would be better off waiting. Unlike the prisoner's dilemma, the bad outcome is one of two equilibria, not the only one.

(e) Verification. `check_L05.py` builds the 4×4 normal form for four parameter sets satisfying $D > r > D/2$ and $R > D$, finds five Nash equilibria, and the two SPNE above, and confirms that (wn, wn) is Nash but not subgame perfect.

Answer. Extensive form: date 1 simultaneous move, date 2 simultaneous move after (n, n) (tree above); normal form: the 4×4 matrix above. There are two pure SPNE: both withdraw at date 1 with payoff (r, r), and both wait at date 1 and withdraw at date 2 with payoff (R, R). In both, the date-2 action is w.

Item 5: Tadelis Exercise 10.2

Foundations · difficulty 2/5

Statement (verbatim from the list, item 5)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.2 "Grim Trigger", p. 215 [PDF 234]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Grim Trigger:** Consider the infinitely repeated game with discount factor $\delta < 1$ of the following variant of the Prisoner's Dilemma:
>
> | Player 1 Player 2 | L | C | R |
> |---|---|---|---|
> | T | 6, 6 | −1, 7 | −2, 8 |
> | M | 7, −1 | 4, 4 | −1, 5 |
> | B | 8, −2 | 5, −1 | 0, 0 |
>
> a. For which values of the discount factor δ can the players support the pair of actions (M, C) played in every period?
>
> b. For which values of the discount factor δ can the players support the pair of actions (T, L) played in every period? Why is your answer different from that for (a)?

(a) What is asked. In the infinitely repeated 3×3 Prisoner's Dilemma variant, find the discount factors for which (M, C) in every period, and then (T, L) in every period, can be sustained, and explain why the two answers differ.

(b) Method. Grim-trigger strategies and the one-stage deviation principle (Tadelis §10.3, pp. 197–200: grim trigger p. 198, Prop. 10.3 p. 198, condition (10.3) p. 200), and the general threshold (10.5), p. 202:

$$\delta \geq \delta^* = \frac{v_i^d - v_i^c}{v_i^d - v_i^*},$$

with v_i^c the cooperative stage payoff, v_i^d the best one-period deviation payoff, and v_i^* the stage Nash payoff used as punishment (`Notas/L05/L05-05`). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17.

(c) Solution. **Notation.** Present value $v_i = \sum_{t \geq 1} \delta^{t-1} u_i(a^t)$ (Tadelis Def. 10.2, p. 193). Best replies are underlined:

	L	C	R
T	6, 6	−1, 7	−2, $\underline{8}$
M	7, −1	4, 4	−1, $\underline{5}$
B	$\underline{8}$, −2	$\underline{5}$, −1	$\underline{0}$, $\underline{0}$

Step 1: the stage equilibrium. Row B beats T ($8 > 6$, $5 > -1$, $0 > -2$) and M ($8 > 7$, $5 > 4$, $0 > -1$) against every column, so B is strictly dominant for player 1. The game is symmetric, so R is strictly dominant for player 2. The unique stage Nash equilibrium is (B, R), with $v_i^* = 0$. It is also

each player's minimax payoff: player 2 can hold player 1 to $\min_c \max_r u_1(r,c) = \min\{8,5,0\} = 0$ by playing R, and B guarantees player 1 at least $\min\{8,5,0\} = 0$.

Step 2: the strategies. For a target profile (a_1, a_2), grim trigger is:

- *cooperation phase*: in period 1, and after any history in which (a_1, a_2) was played in every period, player 1 plays a_1 and player 2 plays a_2;
- *punishment phase*: after any other history, player 1 plays B and player 2 plays R forever.

By Prop. 10.3 this is an SPE if and only if no player gains from a single deviation at a single history. There are two kinds of histories, so there are two checks.

Step 3: punishment phase (both parts). The continuation is (B, R) forever whatever is played today, so a one-period deviation changes only today's payoff. Against R, player 1 gets 0 with B, -1 with M and -2 with T. Deviating loses, for every δ, and player 2 is symmetric.

(a) (M, C)**.** Cooperation phase. Staying gives 4 every period:

$$v_1^{\text{coop}} = \frac{4}{1-\delta}.$$

The best one-period deviation against C is B ($5 > 4 > -1$), and it starts the punishment phase, which pays 0 forever:

$$v_1^{\text{dev}} = 5 + \delta \cdot \frac{0}{1-\delta} = 5.$$

No deviation pays iff

$$\frac{4}{1-\delta} \geq 5 \iff 4 \geq 5 - 5\delta \iff 5\delta \geq 1 \iff \delta \geq \frac{1}{5}.$$

Check with (10.5): $\delta^* = \frac{5-4}{5-0} = \frac{1}{5}$. Player 2's problem is the same by symmetry. So (M, C) in every period is an SPE outcome iff $\delta \geq \frac{1}{5}$.

(b) (T, L)**.** Staying gives $\frac{6}{1-\delta}$. Against L, player 1's best deviation is B (payoff 8), not M (payoff 7). It is followed by 0 forever:

$$\frac{6}{1-\delta} \geq 8 \iff 6 \geq 8 - 8\delta \iff 8\delta \geq 2 \iff \delta \geq \frac{1}{4}.$$

Check: $\delta^* = \frac{8-6}{8-0} = \frac{1}{4}$. The same holds for player 2. So (T, L) in every period is an SPE outcome iff $\delta \geq \frac{1}{4}$.

These thresholds cannot be beaten. The punishment (B, R) forever holds the deviator to 0, his minimax payoff, and he can guarantee 0 in every period by playing B. So in *any* equilibrium (any punishment, even a Nash equilibrium that is not subgame perfect) a deviation from the stationary path is worth at least 5 in (a) and 8 in (b). The thresholds $\frac{1}{5}$ and $\frac{1}{4}$ are therefore necessary as well as sufficient.

Why the answers differ. Write (10.5) as $\delta^* = g/(g+\ell)$, with $g = v^d - v^c$ the one-period gain and $\ell = v^c - v^*$ the per-period loss once punished:

target	g	ℓ	g/ℓ	δ^*
(M, C)	$5 - 4 = 1$	$4 - 0 = 4$	$1/4$	$1/5$
(T, L)	$8 - 6 = 2$	$6 - 0 = 6$	$1/3$	$1/4$

Moving from (M, C) to (T, L) raises the value of cooperating (the loss ℓ rises from 4 to 6), but it raises the temptation more in proportion (the gain g doubles, from 1 to 2). The ratio g/ℓ rises, so more patience is needed. For $\delta \in [\frac{1}{5}, \frac{1}{4})$ the moderate outcome (M, C) can be sustained and the efficient (T, L) cannot.

(d) Interpretation. More ambitious cooperation is harder to sustain when it also creates a bigger temptation. (T, L) is better for both players than (M, C), but at (T, L) each player gives up a lot today by not grabbing the 8. Between moderately patient partners the self-enforcing agreement is the modest one. The same trade-off runs through collusion (Tadelis §10.4): higher collusive prices raise both the gain from undercutting and the value of the cartel.

(e) Verification. `check_C_L05.py` confirms that B and R are strictly dominant, that (B, R) is the unique stage equilibrium with payoff $0 =$ minimax, and that the best deviations are B against C (5) and B against L (8, versus 7 for M). Using exact fractions it tests the cooperation-phase inequality at δ^*, $\delta^* \pm \frac{1}{1000}$ and on the grid $\delta = k/1000$: it holds exactly for $\delta \geq \frac{1}{5}$ in (a) and $\delta \geq \frac{1}{4}$ in (b). The punishment phase has no profitable deviation. Output: "Tadelis 10.2: stage NE (B,R) (0,0); (a) delta* = 1/5, (b) delta* = 1/4; grid k/1000 and +-1/1000 around each cutoff pass". Tadelis prints no answer.

(f) Common mistakes.

- Using M as the deviation from (T, L), because it is "next door". That gives $\frac{7-6}{7-0} = \frac{1}{7}$, which is wrong: the deviation to use is the *best* reply, B.
- Checking only the cooperation phase. Prop. 10.3 needs the punishment phase too; here it holds because (B, R) is a stage equilibrium, and that must be said.
- Reading "(T, L) is better for both" as "(T, L) is easier to sustain". The threshold depends on the ratio of temptation to loss, not on how good the outcome is.

Answer. Unique stage equilibrium (B, R), payoff $(0, 0)$ (also the minimax). With grim trigger (revert to (B, R) forever): (a) (M, C) is sustained iff $\frac{4}{1-\delta} \geq 5$, i.e. $\delta \geq \frac{1}{5}$; (b) (T, L) is sustained iff $\frac{6}{1-\delta} \geq 8$, i.e. $\delta \geq \frac{1}{4}$. Since the punishment is the minimax, no other strategies do better. The answers differ because $\delta^* = g/(g + \ell)$: at (T, L) the temptation doubles ($g = 2$ against 1) while the loss from punishment rises only from 4 to 6.

Item 6: Muñoz-Garcia Ch. 6 Exercise 1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 6)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 1 "Infinitely Repeated Prisoner's Dilemma Game", p. 274 [PDF 285]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider the infinitely repeated version of the following prisoner's dilemma game, where C denotes confess and NC represents not confess (Table 6.1):
>
> **Part (a)** Can players support the cooperative outcome (NC, NC) as a SPNE by using tit-for-tat strategies, whereby players punish deviations from NC in a past period by reverting to the stage-game Nash equilibrium where they both confess (C, C) for just one period, and then returning to cooperation?
>
> **Part (b)** Consider now that players play tit-for-tat, but reverting to the stage-game Nash equilibrium where they both confess (C, C) for two periods, and then players return to cooperation. Can the cooperative outcome (NC, NC) be supported as a SPNE?
>
> **Part (c)** Suppose that players use strategies that punish deviation from cooperation by reverting to the stage-game Nash equilibrium for ten periods before returning to cooperation. Compute the threshold discount factor, δ, above which cooperation can be sustained as a SPNE of the infinitely repeated game.

Player 1 \ Player 2	C	NC
C	2, 2	6, 0
NC	0, 6	3, 3

Table 6.1 Prisoner's dilemma game

(a) What is asked. In the infinitely repeated Prisoner's Dilemma of Table 6.1, can (NC, NC) be an SPE outcome when a deviation is punished by (C, C) for (a) one period, (b) two periods, and (c) for ten periods, what is the threshold δ?

(b) Method. Temporary Nash reversion ("length-k punishment") and the one-stage deviation principle (Tadelis Prop. 10.3, p. 198; grim trigger and (10.3), pp. 198–200; the general threshold (10.5), p. 202). Compare the stream from cooperating with the stream from deviating once and then being punished for k periods; the two streams coincide from period $k+2$ on. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md`.

(c) Solution. Notation. Stage payoffs (player 1 first): $(C,C) = (2,2)$, $(C,NC) = (6,0)$, $(NC,C) = (0,6)$, $(NC,NC) = (3,3)$. Present values $\sum_{t\geq 0} \delta^t u_i^t$, measured from the period in which a player considers deviating. A length-k punishment strategy has two phases:

- *cooperation*: play NC in period 1 and as long as no deviation from the prescribed actions has occurred;
- *punishment*: after a deviation, play C for k periods, then return to cooperation (a deviation during the punishment restarts it).

Step 1: the stage game. C is strictly dominant ($2 > 0$ against C, $6 > 3$ against NC), so (C, C), with $(2, 2)$, is the unique stage equilibrium. The deviation gain is $g = 6 - 3 = 3$; each punishment period costs $\ell = 3 - 2 = 1$.

Step 2: the general condition. Cooperating gives $3 + 3\delta + \dots$. Deviating gives 6 now, 2 in each of the next k periods, then 3 forever. From period $k+1$ on (counting the current period as 0) the streams are equal, so the comparison is over periods $0, \dots, k$:

$$3 + 3\sum_{j=1}^{k} \delta^j \;\geq\; 6 + 2\sum_{j=1}^{k} \delta^j \iff \sum_{j=1}^{k} \delta^j \;\geq\; 3 \iff \frac{\delta(1-\delta^k)}{1-\delta} \;\geq\; 3.$$

In words: the one-period gain, 3, must be covered by k discounted losses of 1 each. Since $\sum_{j=1}^{k} \delta^j < k$ for $\delta < 1$, no punishment of length $k \leq 3$ can ever work.

(a) One period ($k = 1$). The condition is $\delta \geq 3$: impossible for $\delta < 1$. Explicitly, $3 + 3\delta \geq 6 + 2\delta \iff \delta \geq 3$. Cooperation cannot be supported.

(b) Two periods ($k = 2$). The condition is

$$\delta + \delta^2 \geq 3 \iff \delta^2 + \delta - 3 \geq 0.$$

The roots are $\delta = \frac{-1 \pm \sqrt{13}}{2} \approx -2.30,\ 1.30$; the quadratic is negative on $(0, 1)$ (at most $1+1-3 = -1$). Cooperation cannot be supported.

(c) Ten periods ($k = 10$). The condition is

$$\delta + \delta^2 + \dots + \delta^{10} \geq 3.$$

The left side rises from 0 to 10 on $[0, 1]$, so there is exactly one root in $(0, 1)$. Numerically (bisection or a polynomial root-finder),

$$\delta_{10} \approx 0.7627.$$

Cooperation is an SPE outcome iff $\delta \geq \delta_{10}$. For comparison, the thresholds for other lengths are $\delta_4 \approx 0.888$, $\delta_5 \approx 0.834$, $\delta_6 \approx 0.805$, $\delta_8 \approx 0.775$, $\delta_{20} \approx 0.7506$, and grim trigger ($k = \infty$) needs $\frac{\delta}{1-\delta} \geq 3$, i.e. $\delta \geq \frac{3}{4}$ ((10.5): $\frac{6-3}{6-2} = \frac{3}{4}$).

Step 3: the punishment phase. In a punishment period the prescribed action is C, a stage best reply to C. Deviating to NC pays $0 < 2$ today and either leaves the future unchanged or, with the restart rule, lengthens the punishment, which pays $2 < 3$ per period. Either way the deviation loses, for every δ. So the only binding constraint is Step 2's, and the profile is an SPE exactly when that holds.

(d) Interpretation. A punishment deters only if the deviator loses more than he grabs. Here the grab is 3 and each period of punishment costs only 1, so at least four periods are needed, and even then only very patient players are deterred. Longer punishments lower the threshold, with grim trigger as the limit. Short, forgiving punishments are attractive (mistakes are not punished forever), but they need a punishment phase that hurts much more per period than reverting to the Nash equilibrium does here.

(e) Verification. `check_C_L05.py` confirms that (C, C) is the unique stage equilibrium and derives the condition $\sum_{j=1}^{k} \delta^j \geq 3$ symbolically for $k = 1, 2, 3, 4, 10$. It shows (a), (b), and also $k = 3$ fail on the grid $\delta = j/1000$. For (c) it finds the single root in $(0, 1)$, $0.762700984\dots$, and brackets it in exact fractions: the condition fails at 0.762700, holds at 0.762701, fails at 0.76 and holds at 0.77. It also checks that the thresholds decrease in k towards $\frac{3}{4}$. Output: "Munoz-Garcia 6.1: k=1 and k=2 never work (k=3 needs delta=1); k=10: delta >= 0.762701 (book: 0.76, which fails); grim: 3/4; k=4: 0.8882". *Comparison with Muñoz-Garcia's answer* (pp. 275–278): the same

conditions and conclusions for (a) and (b) ($\delta \geq 3$; $\delta^2 + \delta - 3 \geq 0$ with roots -2.3 and 1.3). For (c) the book reports $\delta \geq 0.76$, which is the threshold rounded *down*: at $\delta = 0.76$ the sum is about $2.963 < 3$, so cooperation fails. The correct statement is $\delta \geq 0.7627$. Two slips in the book's text for (a): it says the punisher finds "NC a best response to player i playing NC", meaning C against C; and it calls (NC, NC) the "competitive outcome" where it means the cooperative one. The book does not check the punishment phase.

(f) Common mistakes.

- Calling these strategies "tit-for-tat". True tit-for-tat copies the opponent's last move, which is not the same as "both play C for k periods". Tit-for-tat proper is generally *not* subgame perfect, because after a deviation it requires a player to punish while being punished. The book uses the name loosely.
- Comparing the streams only up to period 1 ("$6 + 2\delta$ versus $3 + 3\delta$") when $k > 1$. All k punishment periods enter the comparison.
- Rounding the threshold to 0.76. A threshold must be rounded up, or stated as the exact root.

Answer. With a k-period reversion to (C, C), cooperation is an SPE iff $\sum_{j=1}^{k} \delta^j \geq 3$ (the punishment phase never binds). (a) $k = 1$: $\delta \geq 3$, impossible. (b) $k = 2$: $\delta^2 + \delta - 3 \geq 0$, root $\frac{-1+\sqrt{13}}{2} \approx 1.30 > 1$, impossible. (c) $k = 10$: $\delta + \delta^2 + \cdots + \delta^{10} \geq 3$, i.e. $\delta \geq \delta_{10} \approx 0.7627$ (the book's 0.76 is slightly too low). Grim trigger would need $\delta \geq \frac{3}{4}$.

Item 7: Tadelis Exercise 10.9

Foundations · difficulty 2/5

Statement (verbatim from the list, item 7)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.9 "Negative Externalities", pp. 218–219 [PDF 237–238]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Negative Externalities:** Two firms are located adjacent to one another, and each imposes an external cost on the other: the detergent that firm 1 uses in its laundry business makes the fish that firm 2 catches in the lake taste funny, and the smoke that firm 2 uses to smoke its caught fish makes the clothes that firm 1 hangs out to dry smell funny. As a consequence each firm's profits are increasing in its own production and decreasing in the production of its neighbor. In particular if q_1 and q_2 are the firms' production levels then their per-period (stage-game) profits are given by $v_1(q_1, q_2) = (30 - q_2)q_1 - q_1^2$ and $v_2(q_1, q_2) = (30 - q_1)q_2 - q_2^2$.
>
> a. Draw the firms' best-response functions and find the Nash equilibrium of the stage-game. How does this compare to the Pareto-optimal stage-game profit levels?
>
> b. For which levels of discount factors can the firms support the Pareto-optimal level of quantities in an infinitely repeated game?

(a) What is asked. Two firms impose external costs on each other: find the stage-game Nash equilibrium and compare it with the Pareto-optimal outputs; then find the discount factors for which the Pareto-optimal outputs are an SPE of the infinitely repeated game.

(b) Method. Grim trigger and the one-stage deviation principle with normalised payoffs (Lecture 5, slides 21, 24–25, 28–29; Tadelis §10.3, pp. 196–202, threshold (10.5), p. 202). The cooperative state is checked against the best one-period deviation followed by the stage Nash equilibrium forever; the punishment state needs no check because it is a stage equilibrium.

(c) Solution. Payoffs: $v_1 = (30 - q_2)q_1 - q_1^2$ and $v_2 = (30 - q_1)q_2 - q_2^2$.

(a) Stage game. The first-order condition of firm 1 is $30 - q_2 - 2q_1 = 0$, so the best responses are

$$q_1(q_2) = \tfrac{30-q_2}{2}, \qquad q_2(q_1) = \tfrac{30-q_1}{2}$$

(downward-sloping lines, each with intercept 15 on its own axis and 30 on the rival's axis, crossing at one point). Solving, $q_1 = q_2 = 10$ is the unique Nash equilibrium, with profits $v_i = (30 - 10) \cdot 10 - 100 = 100$.

Pareto optimum. Let $s = q_1 + q_2$. Then $v_1 + v_2 = 30s - 2q_1q_2 - q_1^2 - q_2^2 = 30s - s^2$, which depends on s only and is maximal at $s = 15$. Any split of 15 maximises joint profit; the symmetric one is $q_1 = q_2 = \frac{15}{2}$, with

$$v_i = (30 - 7.5)\,7.5 - 7.5^2 = 168.75 - 56.25 = 112.5.$$

Compared with Nash, each firm produces less ($7.5 < 10$) and earns more ($112.5 > 100$). Each firm ignores the harm its output does to its neighbour, so the Nash equilibrium overproduces.

(b) Grim trigger. Strategy: produce 7.5 as long as both have; otherwise produce 10 forever.

- *Punishment state.* The continuation is the stage Nash equilibrium whatever happens, and a deviation from 10 cannot raise the stage payoff, so no deviation pays.

- *Cooperative state.* The best deviation against $q_j = 7.5$ is the best response $q_i = \frac{30-7.5}{2} = 11.25$, with profit $v^d = 11.25 \cdot (22.5 - 11.25) = 11.25^2 = 126.5625$.

In normalised payoffs the incentive constraint is

$$112.5 \;\geq\; (1-\delta)\,126.5625 + \delta \cdot 100 \;\Longleftrightarrow\; \delta \;\geq\; \frac{126.5625 - 112.5}{126.5625 - 100} = \frac{14.0625}{26.5625} = \frac{9}{17} \approx 0.529.$$

(This is the general threshold (10.5), $\delta^* = (v^d - v^c)/(v^d - v^*)$.) By symmetry the same holds for firm 2. So the Pareto-optimal outputs $(7.5, 7.5)$ are an SPE outcome of the infinitely repeated game iff $\delta \geq \frac{9}{17}$.

(d) Interpretation. The externality makes cooperation valuable (112.5 against 100) but the deviation temptation is large too (126.5625). Firms must be fairly patient, more than 0.53.

(e) Verification. `check_L05.py` solves the best responses with sympy ($q^* = (10, 10)$, $v = 100$, $\partial^2 v_i / \partial q_i^2 = -2$), shows $v_1 + v_2 = 30s - s^2$, gets 112.5, $q_d = 45/4$, $v^d = 2025/16$ and $\delta^* = 9/17$, and tests the inequality on both sides of 9/17.

Answer. (a) Nash equilibrium $q_1 = q_2 = 10$, profits 100 each; the Pareto-optimal (joint-profit-maximising) symmetric outputs are 7.5 each with profits 112.5: lower output, higher profit. (b) Grim trigger sustains $(7.5, 7.5)$ iff $\delta \geq \frac{9}{17} \approx 0.53$ (best deviation $q = 11.25$, profit 126.5625).

Item 8: Tadelis Exercise 9.3

Foundations · difficulty 3/5

Statement (verbatim from the list, item 8)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.3 "Campaigning Revisited", pp. 187–188 [PDF 206–207]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Campaigning Revisited:** Two political candidates are scheduled to campaign in two states, in one in period $t = 1$ and in the other in $t = 2$. In each state each candidate can choose either a positive campaign that promotes his own agenda (P for player 1, p for player 2) or a negative one that attacks his opponent (N for player 1, n for player 2). Residents of the first-period state do not mind negative campaigns, which are generally effective, and payoffs in this state are given by the following matrix:

Player 1 \ Player 2	p	n
P	2, 2	0, 5
N	5, 0	3, 3

> In the second-period state, residents dislike negative campaigns despite their effectiveness, and the payoffs are given by the following matrix:

Player 1 \ Player 2	p	n
P	4, 4	1, 0
N	0, 1	2, 2

> a. What are the Nash equilibria of each stage-game? Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.
>
> b. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players choose (P, p) in the first stage-game.
>
> c. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (b) survives?
>
> d. Can you find a subgame-perfect equilibrium for this game in which the players play something other than (P, p) or (N, n) in the first stage?

Note (not part of the exercise). As printed, parts (b)–(d) have no solution with $\delta \leq 1$: the best-reply gain from N in stage 1 is 3, while stage 2 can swing a player's payoff by at most $4 - 2 = 2$. The solution proves this and gives the threshold $\delta = 3/2$.

(a) What is asked. For two different stage games played in sequence: find the stage equilibria and the pure SPE at $\delta = 0$; then try to support (P, p) in stage 1 at $\delta = 1$, find the lowest δ for it, and ask whether anything other than (P, p) or (N, n) can be played in stage 1.

(b) Method. Multistage games with a stage-2 game that differs from stage 1 (Tadelis §9.4, pp. 180–184; Prop. 9.2, p. 181: stage-2 play is a stage-2 Nash equilibrium after every history) and the one-stage deviation principle (Thm. 9.1, p. 185). Slides, Lecture 5: slide 10 (last-period play is a stage equilibrium), slides 13–15 (carrot, stick and the condition $\delta \cdot (\text{carrot} - \text{stick}) \geq \text{gain}$).

(c) Solution. Stage games (player 1 chooses rows; best replies underlined):

Stage 1	p	n
P	2, 2	0, $\underline{5}$
N	$\underline{5}$, 0	$\underline{3}$, $\underline{3}$

Stage 2	p	n
P	$\underline{4}$, $\underline{4}$	1, 0
N	0, 1	$\underline{2}$, $\underline{2}$

(a) Stage equilibria and SPE at $\delta = 0$. In stage 1, N beats P against p ($5 > 2$) and against n ($3 > 0$), and n beats p symmetrically, so (N, n) with $(3, 3)$ is the unique equilibrium. In stage 2 there are two pure equilibria, (P, p) with $(4, 4)$ and (N, n) with $(2, 2)$ ((P, n) and (N, p) are not: the player getting 0 or 1 switches). At $\delta = 0$ the future does not matter, so stage 1 is played as a one-shot game, (N, n), and stage 2 is a stage equilibrium after each of the four stage-1 outcomes. The pure SPE are

- stage 1: player 1 plays N, player 2 plays n;
- stage 2: after every history $h \in \{Pp, Pn, Np, Nn\}$ both play (P, p) or both play (N, n), chosen history by history.

That is $2^4 = 16$ SPE, differing only in the choice after each history.

(b)–(c) Supporting (P, p). Try the carrot and stick: play (P, p) in stage 1; in stage 2 play (P, p) after (P, p) and (N, n) after any other history. Stage 2 is a stage equilibrium after every history, so only stage 1 needs checking. Player 1 conforms with $2 + 4\delta$; deviating to N gives 5 now and the stick 2 later:

$$2 + 4\delta \;\geq\; 5 + 2\delta \iff \delta \;\geq\; \tfrac{3}{2}.$$

The same holds for player 2. So the strategies work only for $\delta \geq \frac{3}{2}$. For $\delta \in [0, 1]$ there is no such SPE: the gain from deviating, $5 - 2 = 3$, exceeds the largest possible carrot-minus-stick swing, $\delta(4 - 2) \leq 2$. Using the mixed stage-2 equilibrium (P with probability $\frac{1}{5}$, payoff $\frac{8}{5}$) as the stick does not help: the swing is $4 - \frac{8}{5} = \frac{12}{5}$ and the condition becomes $\delta \geq \frac{5}{4} > 1$.

(d) Other first-stage outcomes. In stage 1, N is a best reply to each action of the rival, and the gain from playing N rather than P is 3 whatever the rival does ($5 - 2$ against p, $3 - 0$ against n). A player who plays P in an SPE on a path with continuation values c^{path} and c^{dev} needs $\delta(c^{\text{path}} - c^{\text{dev}}) \geq 3$, and $c^{\text{path}} - c^{\text{dev}} \leq 4 - 2 = 2$. For $\delta \leq 1$ this is impossible, so neither player ever plays P (or p) in stage 1 of a pure SPE. Hence (P, n), (N, p) and (P, p) are all excluded: the only first-stage outcome of a pure SPE is (N, n), for every $\delta \in [0, 1]$. The answer to (d) is no.

(d) Interpretation. Stage 1's temptation (3) is larger than stage 2 can punish (2), because the stick (N, n) still pays 2 and the carrot pays only 4. Cooperation needs a stage 2 whose best and worst equilibria are far enough apart: $\delta\,(c - s) \geq g$ (slide 15). As printed, (b) and (c) have no answer inside $\delta \leq 1$; the formal threshold is $\frac{3}{2}$.

(e) Verification. `check_L05.py` finds the stage equilibria, confirms $2^4 = 16$, and with the exact one-stage-deviation test shows that for $\delta \in \{0, 0.1, 0.5, 0.9, 1\}$ the only first-stage outcome of a pure SPE is (N, n), that (P, p) becomes feasible at $\delta = \frac{3}{2}$ (and not at 1.49), and that the mixed stick gives $\frac{5}{4}$.

(f) Common mistakes.

- Forcing an answer to (b): at $\delta = 1$ one finds $6 < 7$ for the conforming player, so the strategies above are not an SPE.
- Using the *best* stage-2 outcome as the punishment: the punishment must be the worst equilibrium, (N, n), and it still cannot deter.

Answer. (a) Stage 1: (N, n), $(3, 3)$; stage 2: (P, p) with $(4, 4)$ and (N, n) with $(2, 2)$; at $\delta = 0$ there are 16 pure SPE (stage 1 (N, n), and a stage-2 equilibrium chosen after each of the four histories). (b)–(c) (P, p) in stage 1 with reward (P, p) and punishment (N, n) requires $\delta \geq \frac{3}{2}$, so there is no such SPE for $\delta \leq 1$. (d) No: for $\delta \leq 1$ the only first-stage outcome of a pure SPE is (N, n).

Item 9: Cabral Exercise 7.12

Exam level · difficulty 2/5

Statement (verbatim from the list, item 9)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 7 Challenging exercises, Exercise 7.12 "Ad games", p. 181 [PDF 193]. Answer in the source: none in the book. Reproduced verbatim.

> **7.12. Ad games.** Two firms must simultaneously choose their advertising budget; their options are H or L. Payoffs are as follows: if both choose H, then each gets 5; if both choose L, then each gets 4; if firm 1 chooses H and firm 2 chooses L, then firm 1 gets 8 and firm 2 gets 1; conversely, if firm 2 chooses H and firm 1 chooses L, then firm 2 gets 8 and firm 1 gets 1.
>
> (a) Determine the Nash equilibria of the one-shot game.
>
> (b) Suppose the game is indefinitely repeated and that the relevant discount factor is $\delta = .8$. Determine the optimal symmetric equilibrium.
>
> (c) (challenge question) Now suppose that, for the first 10 periods, firm payoffs are twice the values represented in the above table. What is the optimal symmetric equilibrium?

Note (not part of the exercise). With the printed payoffs (H, H) gives 5 each and (L, L) only 4 each, so H is dominant and also efficient. The solution works the exercise as printed.

(a) What is asked. Find the Nash equilibria of a symmetric 2×2 advertising game; find the optimal symmetric equilibrium of the infinitely repeated game at $\delta = 0.8$; and again when payoffs are doubled in the first ten periods.

(b) Method. Nash equilibrium of the stage game, stage-game repetition as an SPE, and the one-stage deviation test for grim trigger with normalised payoffs (Lecture 5, slides 10, 21, 24–29; Tadelis §10.3, pp. 196–202). The best symmetric equilibrium is found by comparing each period's best symmetric stage payoff with what the equilibrium conditions allow.

(c) Solution. Stage game (firm 1 chooses rows, best replies underlined):

	H	L
H	$\underline{5}, \underline{5}$	$\underline{8}, 1$
L	$1, \underline{8}$	$4, 4$

(a) H strictly dominates L: $5 > 1$ against H and $8 > 4$ against L (the game is symmetric). The unique Nash equilibrium is (H, H) with $(5, 5)$.

(b) Unlike the prisoner's dilemma, the stage equilibrium is also the best symmetric outcome: (H, H) gives 5 each, (L, L) only 4. Three facts settle the question.

- (H, H) *in every period is an SPE* for every δ: it is a stage equilibrium after every history, so there is no profitable one-stage deviation (slide 10).
- *It gives each firm* 5 *in every period*, the largest payoff either firm can get in a symmetric outcome: the symmetric outcomes are (H, H) with 5 and (L, L) with 4, and alternating (H, L), (L, H) gives $\frac{8+1}{2} = 4.5$ on average. No equilibrium can give a symmetric payoff above 5.

- *Cooperating on (L, L) is not even an equilibrium.* Grim trigger with reversion to (H, H) needs $4 \geq (1-\delta)\cdot 8 + \delta \cdot 5$, i.e. $\delta \geq \frac{4}{3} > 1$: the "punishment" pays $5 > 4$.

So at $\delta = 0.8$ the optimal symmetric equilibrium is (H, H) forever, with normalised payoff 5 per firm.

(c) In periods 1–10 all payoffs are doubled: $(H, H) = (10, 10)$, $(L, L) = (8, 8)$, $(H, L) = (16, 2)$, and from period 11 the original table applies. H is still strictly dominant in every period, (H, H) is still the unique stage equilibrium, and it is still the best symmetric outcome in every period ($10 > 8$, $5 > 4$). The same argument applies, so the optimal symmetric equilibrium is again (H, H) in every period. Its normalised value is

$$(1-\delta)\Big[\,10\,\tfrac{1-\delta^{10}}{1-\delta} + 5\,\tfrac{\delta^{10}}{1-\delta}\Big] = 10 - 5\delta^{10} = 10 - 5\cdot 0.8^{10} \approx 9.463.$$

Front-loading the payoffs raises the value but changes no incentive: (L, L) is dominated period by period.

(d) Interpretation. The exercise reads like a collusion problem, but with these payoffs there is nothing to collude on: mutual high advertising is both the equilibrium and the efficient symmetric outcome. For (L, L) to be a cooperative outcome one needs $u(L, L) > u(H, H)$, as in the prisoner's dilemma.

(e) Verification. `check_L05.py` computes the unique equilibrium (H, H) with H dominant, the grim-trigger threshold $\frac{4}{3}$, the best symmetric stage payoff 5 and total payoff 10, and the value of part (c) with exact fractions: average payoff 9.4631.

(f) Common mistakes.

- Treating the game as a prisoner's dilemma and answering "(L, L) for $\delta \geq \ldots$". With $5 > 4$ the check fails for every $\delta < 1$.
- In (c), computing a threshold from the doubled payoffs: doubling does not change which outcome is best in a period, so the answer is unchanged.

Answer. (a) (H, H) with $(5, 5)$ is the unique Nash equilibrium (H dominant). (b) At $\delta = 0.8$ the optimal symmetric equilibrium is (H, H) in every period, payoff 5 each; (L, L) cannot be sustained (it would need $\delta \geq \frac{4}{3}$). (c) Same equilibrium, (H, H) in every period; average payoff $10 - 5 \cdot 0.8^{10} \approx 9.46$.

Item 10: Tadelis Exercise 10.3

Exam level · difficulty 3/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.3 "Not-So-Grim Trigger", pp. 215–216 [PDF 234–235], with its footnote 12 (p. 216). Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim. (Here T is the length of the punishment, not a payoff.)

> **Not-So-Grim Trigger:** Consider the infinitely repeated Prisoner's Dilemma with discount factor $\delta < 1$ described by the following matrix:
>
> | Player 1 \ Player 2 | m | f |
> |---|---|---|
> | M | 4, 4 | -1, 5 |
> | F | 5, -1 | 1, 1 |
>
> Instead of using grim-trigger strategies to support a pair of actions (a_1, a_2) other than (F, f) as a subgame-perfect equilibrium, assume that the players wish to choose a less draconian punishment called a "length-T punishment" strategy. If there is a deviation from (a_1, a_2) then the players will play (F, f) for T periods and then resume playing (a_1, a_2). Let δ_T be the critical discount factor so that if $\delta > \delta_T$ then the adequately defined strategies will implement the desired path of play with length-T punishment as the threat.
>
> a. Let $T = 1$. What is the critical value δ_1 to support the pair of actions (M, m) played in every period?
>
> b. Let $T = 2$. What is the critical value δ_T to support the pair of actions (M, m) played in every period?[12]
>
> c. Compare the two critical values in (a) and (b). How do they differ and what is the intuition for this?
>
> [12] Hint: You should encounter an equation of the form $a\delta^3 - (a+1)\delta + 1 = 0$, for which it is easy to see that $\delta = 1$ is a root. In this case you know that the equation can be written in the form $(\delta - 1)(a\delta^2 + a\delta - 1) = 0$, and you can solve for the other relevant root of the cubic equation.

(a) What is asked. In Tadelis's repeated Prisoner's Dilemma, find the critical discount factor for (M, m) every period under a length-T punishment, (a) for $T = 1$ and (b) for $T = 2$, and (c) compare the two and explain the difference.

(b) Method. Length-T punishments checked with the one-stage deviation principle (Prop. 10.3, p. 198), against the grim-trigger benchmark (10.3), p. 200 ($\delta \geq \frac{1}{4}$ in this game) and (10.5), p. 202. The cooperation-phase test compares the two streams only over the periods in which they differ. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md` and `L05-05-critical-discount-factor.md`.

(c) Solution. Notation. Stage payoffs: $(M, m) = (4, 4)$, $(M, f) = (-1, 5)$, $(F, m) = (5, -1)$, $(F, f) = (1, 1)$. F and f are strictly dominant ($5 > 4$, $1 > -1$), so (F, f) with $(1, 1)$ is the unique stage equilibrium. Strategies with a length-T punishment:

- *cooperation phase*: play M (player 1) and m (player 2);
- *punishment phase*: after any deviation from the prescribed actions, play (F, f) for T periods, then return to the cooperation phase (a deviation during the punishment restarts it).

Present values are measured from the period of the deviation (period 0).

Step 1: the general cooperation-phase condition. Cooperating gives 4 in periods $0, 1, \ldots, T$; deviating gives 5 in period 0 and 1 in periods $1, \ldots, T$. From period $T+1$ on both streams give 4 forever. So no deviation pays iff

$$4 + 4\sum_{j=1}^{T} \delta^j \;\geq\; 5 + \sum_{j=1}^{T} \delta^j \iff 3\sum_{j=1}^{T} \delta^j \;\geq\; 1.$$

The one-period gain is $5 - 4 = 1$ and each punishment period costs $4 - 1 = 3$.

Step 2: the punishment phase. The prescribed action F is a stage best reply to f. Deviating to M pays $-1 < 1$ today and does not shorten the punishment (with the restart rule it lengthens it, and punishment periods pay $1 < 4$). So no one-stage deviation pays in the punishment phase, for any δ and any T. Only Step 1 binds.

(a) $T = 1$. Using present values, as the book's hint suggests:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta \cdot 1 + \frac{4\delta^2}{1-\delta}.$$

Multiplying by $1 - \delta > 0$: $4 \geq 5(1-\delta) + \delta(1-\delta) + 4\delta^2 = 5 - 4\delta + 3\delta^2$, i.e. $3\delta^2 - 4\delta + 1 \leq 0$, i.e. $(3\delta - 1)(\delta - 1) \leq 0$. With $\delta < 1$ this is $\delta \geq \frac{1}{3}$. Step 1 gives the same: $3\delta \geq 1$. So

$$\delta_1 = \frac{1}{3}.$$

(b) $T = 2$. Present values:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta + \delta^2 + \frac{4\delta^3}{1-\delta}.$$

Multiplying by $1 - \delta$: $4 \geq 5(1-\delta) + (\delta + \delta^2)(1-\delta) + 4\delta^3 = 5 - 4\delta + 3\delta^3$, i.e.

$$3\delta^3 - 4\delta + 1 \;\leq\; 0.$$

This is the hint's cubic $a\delta^3 - (a+1)\delta + 1$ with $a = 3$. Since $\delta = 1$ is a root,

$$3\delta^3 - 4\delta + 1 = (\delta - 1)(3\delta^2 + 3\delta - 1).$$

For $\delta < 1$ the first factor is negative, so the inequality holds iff $3\delta^2 + 3\delta - 1 \geq 0$ (the same as Step 1: $3(\delta + \delta^2) \geq 1$). The positive root is

$$\delta_2 = \frac{-3 + \sqrt{9 + 12}}{6} = \frac{\sqrt{21} - 3}{6} \approx 0.2638,$$

and (M, m) in every period is sustained iff $\delta \geq \delta_2$. (The other root, $\frac{-3-\sqrt{21}}{6} < 0$, is irrelevant.)

(c) Comparison.

$$\delta_1 = \tfrac{1}{3} \approx 0.333 \;>\; \delta_2 \approx 0.264 \;>\; \delta_\infty = \tfrac{1}{4},$$

where δ_∞ is the grim-trigger threshold ((10.3): $\frac{5-4}{5-1} = \frac{1}{4}$). A longer punishment costs the deviator more: with $T = 1$ the loss is 3δ, with $T = 2$ it is $3(\delta + \delta^2)$, with grim trigger $3\delta/(1-\delta)$. The gain is 1 in every case. A bigger loss covers the same gain at a lower δ, so the critical value falls as T grows, and it falls by less and less because later punishment periods are discounted more (here $\delta_1 - \delta_2 \approx 0.070$ but $\delta_2 - \delta_\infty \approx 0.014$).

(d) Interpretation. Punishments need not be eternal. In this game the stage gain from cheating is small relative to the loss of a punished period (1 against 3), so even a one-period punishment works for moderately patient players, and two periods already get close to what grim trigger achieves. Short punishments are less costly if a "deviation" was in fact a mistake or a misread signal, since cooperation resumes after T periods. The price is a higher patience requirement.

(e) Verification. `check_C_L05.py` derives the condition $3\sum_{j=1}^{T}\delta^j \geq 1$ symbolically for $T = 1, 2, 5$. It checks $\delta_1 = \frac{1}{3}$ at the threshold and at $\pm\frac{1}{1000}$. It rebuilds the book's cubic from the present values ($3\delta^3 - 4\delta + 1$), checks the factorization and the root $\frac{\sqrt{21}-3}{6} = 0.263763$, brackets that root to 10^{-6} in exact fractions, tests the grid $\delta = j/1000$, and checks the ordering $\frac{1}{3} > \delta_2 > \frac{1}{4}$. Output: "Tadelis 10.3: delta_1 = 1/3; delta_2 = (sqrt(21) - 3)/6 = 0.263763; grim 1/4". Tadelis prints no answer.

(f) Common mistakes.

- Forgetting that cooperation *resumes*: writing the deviation stream as $5 + \delta + \delta^2$ and stopping there, or as $5 + \frac{\delta}{1-\delta}$ (which is grim trigger).
- Keeping the root $\delta = 1$ of the cubic as the answer. It is the trivial root that the hint tells you to factor out; the threshold is the root of the quadratic factor.
- Saying that a shorter punishment is "more lenient, so cooperation is easier". It is the opposite: a weaker threat needs more patience.

Answer. With a length-T reversion to (F, f), (M, m) forever is an SPE iff $3\sum_{j=1}^{T}\delta^j \geq 1$ (the punishment phase never binds). (a) $T = 1$: $\delta_1 = \frac{1}{3}$. (b) $T = 2$: $3\delta^3 - 4\delta + 1 = (\delta - 1)(3\delta^2 + 3\delta - 1) \leq 0$, so $\delta_2 = \frac{\sqrt{21}-3}{6} \approx 0.264$. (c) $\delta_1 > \delta_2 > \frac{1}{4}$ (grim trigger): a longer punishment makes deviating costlier for the same one-period gain, so less patience is needed, with diminishing effect as T grows.

Item 11: Gibbons Problem 2.10

Exam level · difficulty 3/5

Statement (verbatim from the list, item 11)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §2.3 Problems, Problem 2.10, p. 134 [PDF 141]. Answer in the source: none in the book. Reproduced verbatim.

> 2.10. The accompanying simultaneous-move game is played twice, with the outcome of the first stage observed before the second stage begins. There is no discounting. The variable x is greater than 4, so that $(4, 4)$ is not an equilibrium payoff in the one-shot game. For what values of x is the following strategy (played by both players) a subgame-perfect Nash equilibrium?
>
> > Play Q_i in the first stage. If the first-stage outcome is (Q_1, Q_2), play P_i in the second stage. If the first-stage outcome is (y, Q_2) where $y \neq Q_1$, play R_i in the second stage. If the first-stage outcome is (Q_1, z) where $z \neq Q_2$, play S_i in the second stage. If the first-stage outcome is (y, z) where $y \neq Q_1$ and $z \neq Q_2$, play P_i in the second stage.

	P_2	Q_2	R_2	S_2
P_1	2, 2	x, 0	-1, 0	0, 0
Q_1	0, x	4, 4	-1, 0	0, 0
R_1	0, 0	0, 0	0, 2	0, 0
S_1	0, -1	0, -1	-1, -1	2, 0

(a) What is asked. For which $x > 4$ is the stated strategy (play Q_i in stage 1, then a continuation that depends on who deviated) a subgame-perfect equilibrium of the twice-played 4×4 game, with no discounting?

(b) Method. Two-stage repeated game with player-specific punishments (Gibbons §2.3.A, pp. 82–88; Tadelis §9.4, pp. 180–184, Thm. 9.1, p. 185). Second-stage play must be a stage Nash equilibrium after every history; a first-stage deviation is checked one stage at a time. Lecture 5, slides 10–15.

(c) Solution. **Step 1: stage equilibria.** Best replies (underlined, with $x > 4$):

	P_2	Q_2	R_2	S_2
P_1	$\underline{2}$, $\underline{2}$	$\underline{x}$, 0	-1, 0	0, 0
Q_1	0, $\underline{x}$	4, 4	-1, 0	0, 0
R_1	0, 0	0, 0	$\underline{0}$, $\underline{2}$	0, 0
S_1	0, -1	0, -1	-1, -1	$\underline{2}$, $\underline{0}$

Player 1's best replies are P_1 to P_2 (2), P_1 to Q_2 (x), R_1 to R_2 (0) and S_1 to S_2 (2); player 2's are P_2 to P_1 (2), P_2 to Q_1 (x), R_2 to R_1 (2), S_2 to S_1 (0). The pure equilibria are

$$(P_1, P_2) \to (2, 2), \qquad (R_1, R_2) \to (0, 2), \qquad (S_1, S_2) \to (2, 0).$$

(Q_1, Q_2) is not one because $x > 4$: either player gains by switching to P.

Step 2: the second stage. The strategy prescribes (P_1, P_2) after (Q_1, Q_2), (R_1, R_2) after a deviation by player 1 only, (S_1, S_2) after a deviation by player 2 only, and (P_1, P_2) after a double deviation. All four are stage equilibria and nothing follows stage 2, so there is no profitable deviation there. Player 1 is punished by (R, R) (payoff 0, his worst equilibrium), and player 2 by (S, S) (payoff 0, her worst).

Step 3: the first stage, player 1. On the path he gets $4 + 2 = 6$. Against Q_2 his deviations give: P_1 pays x, R_1 pays -1, S_1 pays 0; after any of them stage 2 is (R, R) and pays him 0. The best deviation is P_1 with $x + 0 = x$. So he conforms iff

$$6 \geq x.$$

Player 2 is symmetric: against Q_1 the best deviation is P_2 (payoff x), followed by (S, S) and 0; she conforms iff $6 \geq x$. A double deviation is not a one-stage deviation by either player.

Result. The strategy is an SPE iff $4 < x \leq 6$ (the lower bound is only the assumption $x > 4$).

(d) Interpretation. The reward (P, P) is worth 2 to each; the stick is worth 0. The swing, 2, must cover the deviation gain $x - 4$. Player-specific sticks matter: (R, R) hurts only player 1 and (S, S) only player 2. With a single stick equal to (P, P) itself (no punishment) the condition would be $6 \geq x + 2$, impossible for $x > 4$.

(e) Verification. `check_L05.py` lists the pure stage equilibria (P, P), (R, R), (S, S) and, with the exact one-stage-deviation test, finds that (Q, Q) is a first-stage SPE outcome for $x \in \{4.5, 5, 6\}$ and not for $x \in \{6.1, 7\}$: precisely when $x \leq 6$.

Answer. The strategy is an SPE iff $4 < x \leq 6$. Stage 2 is a stage equilibrium after every history: (P, P) on the path, (R, R) after player 1 deviates, (S, S) after player 2 deviates. In stage 1 the best deviation gives x and then 0, against $4 + 2 = 6$ on the path.

Item 12: Tadelis Exercise 9.2

Exam level · difficulty 3/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.2 "Centipedes Revisited", p. 187 [PDF 206]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Centipedes Revisited:** Two players are playing two consecutive games. First, they play the Centipede Game described in Figure 9.2. After the Centipede Game they play the following coordination game:

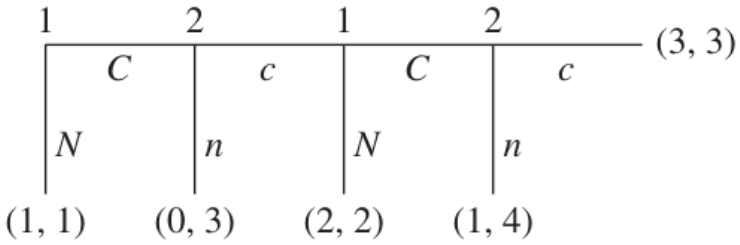


FIGURE 9.2 The Centipede Game.

Player 1	Player 2	a	b
	A	1, 1	0, 0
	B	0, 0	3, 3

a. What are the Nash equilibria of each stage-game?

b. How many pure strategies does each player have in the multistage game?

c. Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.

d. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players receive the payoffs (2, 2) in the first stage-game.

e. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (d) survives?

f. For δ greater than the value you found in (e), are there other outcomes of the first-stage Centipede Game that can be supported as part of a subgame-perfect equilibrium?

(a) What is asked. A centipede game followed by a coordination game: find the stage equilibria, count the strategies, find the SPE at $\delta = 0$, support the outcome $(2, 2)$ in the centipede at $\delta = 1$, find the lowest δ for it, and ask which other centipede outcomes can be supported.

(b) Method. Multistage games (Tadelis §9.4, pp. 180–184; Prop. 9.2, p. 181; one-stage deviation principle, Thm. 9.1, p. 185). Stage 2 is a coordination game with two pure equilibria, so it supplies a carrot (B, b) and a stick (A, a). Lecture 5, slides 10–15 (the same construction on a simultaneous stage 1), with backward induction on the tree (slide 14).

(c) Solution. The centipede (Figure 9.2). Nodes: 1 chooses N (stop, payoff $(1,1)$) or C; then 2 chooses n ($(0,3)$) or c; then 1 chooses N ($(2,2)$) or C; then 2 chooses n ($(1,4)$) or c ($(3,3)$). Write the five terminal outcomes as $o_1 = (1,1)$, $o_2 = (0,3)$, $o_3 = (2,2)$, $o_4 = (1,4)$, $o_5 = (3,3)$. The second game is

	a	b
A	$\underline{1}$, $\underline{1}$	0, 0
B	0, 0	$\underline{3}$, $\underline{3}$

(a) Stage equilibria. *Coordination game:* (A,a) with $(1,1)$ and (B,b) with $(3,3)$, plus a mixed equilibrium (A with probability $\frac{3}{4}$, payoff $\frac{3}{4}$), which we ignore because the exercise asks for pure strategies. *Centipede:* backward induction gives n at the last node ($4 > 3$), N at node 3 ($2 > 1$), n at node 2 ($3 > 2$) and N at node 1 ($1 > 0$): the unique SPE is $(N, N;\, n, n)$ with outcome $(1,1)$. Nash equilibria: every profile in which player 1 plays N at node 1 and player 2 plays n at node 2 (the later choices are off the path), so four profiles, all with outcome $(1,1)$. A profile with c at node 2 is not Nash, because player 1 would then go C and stop at node 3 with $2 > 1$.

(b) Strategies. Player 1 chooses an action at each of his two centipede nodes ($2^2 = 4$ ways) and a stage-2 action after each of the five centipede outcomes ($2^5 = 32$ ways): $4 \cdot 32 = 128$ pure strategies. Player 2 likewise, 128.

(c) SPE at $\delta = 0$. Stage 2 is a stage equilibrium after each outcome, and stage 1 is played without regard to the future, hence by backward induction: $(N, N;\, n, n)$. The SPE are those with this centipede play and, after each of the five outcomes o_k, either (A,a) or (B,b): $2^5 = 32$ pure SPE.

(d) $(2,2)$ at $\delta = 1$. Strategies: player 1 plays C at node 1 and N at node 3; player 2 plays c at node 2 and n at node 4. Stage 2: (B,b) after o_3 (the path) and (A,a) after every other outcome. Stage 2 is a stage equilibrium after every history. In stage 1, one-stage deviations (values are stage-1 payoff plus $\delta\times$ stage-2 payoff):

node	choice	conform	deviate
1 (player 1)	C	$2 + 3\delta$	N: $1 + \delta$
2 (player 2)	c	$2 + 3\delta$	n: $3 + \delta$
3 (player 1)	N	$2 + 3\delta$	C, then n: $1 + \delta$
4 (player 2, off path)	n	$4 + \delta$	c: $3 + \delta$

Nodes 1, 3 and 4 hold for every $\delta \geq 0$. Node 2 gives $2 + 3\delta \geq 3 + \delta$ iff $\delta \geq \frac{1}{2}$, so the strategies are an SPE at $\delta = 1$ and indeed for all $\delta \geq \frac{1}{2}$.

(e) Lowest δ. Node 2 is binding: player 2 gains $3 - 2 = 1$ by stopping, and the largest swing in stage 2 is $3 - 1 = 2$, so $\delta \cdot 2 \geq 1$. No other continuation does better, since (B,b) and (A,a) are the best and worst stage-2 equilibria. The lowest value is $\delta = \frac{1}{2}$; below it the only outcome of a pure SPE is $(1,1)$ (checked by script for $\delta = 0.499$).

(f) Other outcomes for $\delta \geq \frac{1}{2}$. Reward (B,b) after the target outcome only, (A,a) elsewhere, and let play off the path be backward induction. The binding deviation is:

target outcome	binding deviation	condition
$(1,1)$	none	every δ
$(0,3)$	player 1 at node 1 stops: $1+\delta$	$0+3\delta \geq 1+\delta$
$(2,2)$	player 2 at node 2 stops: $3+\delta$	$2+3\delta \geq 3+\delta$
$(1,4)$	player 1 at node 3 stops: $2+\delta$	$1+3\delta \geq 2+\delta$
$(3,3)$	player 2 at node 4 stops: $4+\delta$	$3+3\delta \geq 4+\delta$

Every condition reduces to $\delta \geq \frac{1}{2}$. Hence for $\delta \geq \frac{1}{2}$ *every* outcome of the centipede, including the efficient $(3,3)$, can be a first-stage SPE outcome; for $\delta < \frac{1}{2}$ only $(1,1)$ can.

(d) Interpretation. The coordination game turns the centipede's unravelling around: each stopping gain in the tree is at most 1, and the future carrot is worth 2δ more than the stick. Once $\delta \geq \frac{1}{2}$ the last-node temptation (4 against 3) is also covered, so even full cooperation is supported.

(e) Verification. `check_L05.py` finds the four Nash equilibria of the centipede (all with outcome $(1,1)$), the counts 128 and 32, and, by backward induction with every assignment of $(A,a)/(B,b)$ to the five outcomes, that the set of SPE outcomes is $\{(1,1)\}$ for $\delta < \frac{1}{2}$ and all five outcomes for $\delta = \frac{1}{2}$ and $\delta = 1$; for the strategies in (d), $(2,2)$ is the SPE outcome at $\delta \geq \frac{1}{2}$ and not at 0.49.

Answer. (a) Centipede: unique SPE $(N,N;n,n)$, outcome $(1,1)$; four Nash equilibria, all with that outcome. Coordination game: (A,a), (B,b). (b) $4 \cdot 2^5 = 128$ pure strategies per player. (c) $2^5 = 32$ pure SPE: $(N,N;n,n)$ plus (A,a) or (B,b) after each of the five outcomes. (d) C,N / c,n with (B,b) after $(2,2)$ and (A,a) otherwise. (e) $\delta = \frac{1}{2}$. (f) Yes: for $\delta \geq \frac{1}{2}$ every outcome of the centipede can be supported, with the same $\frac{1}{2}$ threshold.

Item 13: Tadelis Exercise 9.4

Exam level · difficulty 3/5

Statement (verbatim from the list, item 13)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.4 "Online Gaming", p. 188 [PDF 207]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Online Gaming:** Consider a two-stage game between two firms that produce online games. In the first stage, they play a Cournot competition game (each chooses a quantity q_i) with demand function $p = 100 - q$ and zero marginal production costs ($c_i(q_i) = 0$ for $i = 1, 2$). In the second stage, after observing the pair (q_1, q_2) and after profits have been distributed, the players play a simultaneous-move "access" game in which each can open its platform to allow players on the other platform to play online with players on its platform (O for player 1, o for player 2), or choose to keep its platform noncompatible (N for player 1, n for player 2), in which case each platform's players can play only with others on the same platform. If they choose (N, n) then second-stage payoffs are $(0, 0)$. If only one firm chooses to open its platform, it bears a cost of -10 with no benefit, since the other firm did not allow open access. Finally if both firms choose (O, o) then each firm gets many more eyeballs for advertising, and payoffs for each are 2500. Both players use the same discount factor δ to discount future payoffs.
>
> a. Find the unique Nash equilibrium in the first-stage Cournot game and all of the pure-strategy Nash equilibria of the second-stage access game. Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.
>
> b. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly (total profit–maximizing) quantities and split them equally (a symmetric equilibrium).
>
> c. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (b) survives?
>
> d. Now let $\delta = 0.4$. Can you support a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly quantities and split them equally? If not, what are the highest profits that the firms can make in a symmetric equilibrium?

Note (not part of the exercise). With the printed payoffs the threshold in (c) is $1/16$, so the answer to (d) at $\delta = 0.4$ is "yes" and the "if not" branch never arises.

(a) What is asked. Two firms play a Cournot game and then an access game: find the equilibria and the SPE at $\delta = 0$; support the equal split of the monopoly output in stage 1 at $\delta = 1$; find the lowest δ for it; and decide whether it is supportable at $\delta = 0.4$.

(b) Method. Carrot and stick over a continuous first stage (Tadelis §9.4, pp. 180–184, Prop. 9.2, p. 181; Thm. 9.1, p. 185). Stage 2 supplies the reward (O, o) and the stick (N, n); the stage-1 incentive constraint compares the deviation gain with δ [carrot − stick] (Lecture 5, slide 15, $\delta \geq \frac{1}{2}$ in the slide example).

(c) Solution. **(a) Equilibria.** *Stage 1.* Profit $\pi_i = (100 - q_i - q_j)q_i$, best response $q_i = \frac{100 - q_j}{2}$; the unique equilibrium is $q_1 = q_2 = \frac{100}{3}$, with price $\frac{100}{3}$ and profit $\frac{10000}{9} \approx 1111.1$ each. *Stage 2* (best replies underlined):

	o	n
O	$\underline{2500},\ \underline{2500}$	$-10,\ 0$
N	$0,\ -10$	$\underline{0},\ \underline{0}$

The pure equilibria are (O, o) with $(2500, 2500)$ and (N, n) with $(0, 0)$. ((O, n) and (N, o) are not: the firm that opened would rather close.) The mixed equilibrium, O with probability $\frac{1}{251}$, also pays 0.

SPE at $\delta = 0$. Stage 1 is a one-shot game, so $q_1 = q_2 = \frac{100}{3}$; stage 2 must be a stage equilibrium after every history (q_1, q_2): any function from histories to $\{(O, o), (N, n)\}$. A history is a pair of quantities, so there are infinitely many such SPE, all with the same outcome in stage 1.

(b) Monopoly quantities at $\delta = 1$. Total profit $(100 - Q)Q$ is maximal at $Q = 50$, price 50, profit 2500; split equally, $q_i = 25$ and $\pi_i = 1250$. Strategies: each firm produces 25 in stage 1; in stage 2 both play (O, o) after the history $(25, 25)$ and (N, n) after every other history. Stage 2 is a stage equilibrium after every history.

(c) Lowest δ. The best deviation against $q_j = 25$ is the best response $q_i = \frac{100-25}{2} = 37.5$, with profit $37.5 \cdot (100 - 62.5) = 37.5^2 = 1406.25$, followed by the stick (N, n) worth 0. Conforming gives $1250 + 2500\delta$:

$$1250 + 2500\,\delta \ \geq\ 1406.25 + 0 \cdot \delta \iff \delta \ \geq\ \frac{156.25}{2500} = \frac{1}{16} = 0.0625.$$

The stick 0 is the worst stage-2 payoff (pure or mixed), so no punishment does better. The threshold is $\delta = \frac{1}{16}$.

(d) $\delta = 0.4$. Since $0.4 > \frac{1}{16}$, *yes*: the strategies of (b) are an SPE with payoffs $1250 + 0.4 \cdot 2500 = 2250$ per firm. The question's "if not" branch does not arise. For completeness, below $\frac{1}{16}$ the lowest symmetric quantity that can be sustained solves $\left(\frac{100-q}{2}\right)^2 - (100 - 2q)q = 2500\delta$, i.e. $q_{\min}(\delta) = \frac{100}{3}\left(1 - \sqrt{\delta}\right)$. It equals the Cournot quantity at $\delta = 0$ and the monopoly split 25 at $\delta = \frac{1}{16}$.

(d) Interpretation. Stage 2 is a very strong carrot: moving from (N, n) to (O, o) is worth 2500, twice the monopoly profit in stage 1. A very small weight on the future ($\frac{1}{16}$) is enough to keep the firms from undercutting.

(e) Verification. `check_L05.py` solves the Cournot equilibrium ($\frac{100}{3}$, profit $\frac{10000}{9}$), the monopoly (50, 2500), the deviation ($\frac{75}{2}$, $\frac{5625}{4}$), the stage-2 equilibria and $\delta^* = \frac{1}{16}$ (both sides tested, 0.4 included), and the formula for $q_{\min}(\delta)$.

(f) Common mistakes.

- Treating -10 as the stick: it is the payoff of an off-equilibrium profile. The stick is the equilibrium (N, n) with payoff 0.
- Counting stage 1 deviations by quantity only: after *any* history other than $(25, 25)$ the firms play (N, n), including after a deviation by only one of them.

Answer. (a) Cournot $q_i = \frac{100}{3}$ (profit $\frac{10000}{9}$); stage 2 equilibria (O, o) and (N, n); at $\delta = 0$ the SPE have $q_i = \frac{100}{3}$ and a stage-2 equilibrium chosen after every history. (b) $q_i = 25$, then (O, o) after $(25, 25)$ and (N, n) otherwise. (c) $\delta \geq \frac{1}{16}$ (best deviation 37.5, profit 1406.25). (d) Yes at $\delta = 0.4$ (it exceeds $\frac{1}{16}$).

Item 14: Muñoz-Garcia Ch. 6 Exercise 6

Exam level · difficulty 3/5

Statement (verbatim from the list, item 14)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 6 "Collusion in Donations in an Infinitely Repeated Public Good Game", pp. 292–293 [PDF 303–304]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it; "Grim-Triger" is the book's spelling).

> Consider a public good game between two players, A and B. The utility function of every player $i = \{A, B\}$ is
>
> $$u_i(g_i) = (w - g_i)^{1/2} \left[m\left(g_i + g_j\right)\right]^{1/2}$$
>
> where the first term, $w - g_i$, represents the utility that the player receives from money (i.e., the amount of his wealth, w, not contributed to the public good). The second term indicates the utility he obtains from aggregate contributions to the public good, $g_i + g_j$, where $m \geq 0$ denotes the return from aggregate contributions. For simplicity, assume that both players have equal wealth levels w.
>
> (a) Find every player i's best response function, $g_i(g_j)$. Interpret.
>
> (b) Find the Nash Equilibrium of this game, g^*.
>
> (c) Find the socially optimal contribution level, g^{SO}, that is, the donation that each player should contribute to maximize their joint utilities. Compare it with the Nash equilibrium of the game. Which of them yields the highest utility for every player i?
>
> (d) Consider the infinitely repeated version of the game. Show under which values of players' discount factor δ you can sustain cooperation, where cooperation is understood as contributing the socially optimal amount (the one that maximizes joint utilities) you found in part (c). For simplicity, you can consider a "Grim-Triger strategy" where every player starts cooperating (that is, donating g^{SO} to the public good), and continues to do so if both players cooperated in all previous periods. However, if one or both players deviate from g^{SO}, he reverts to the Nash equilibrium of the unrepeated game, g^*, thereafter.

(a) What is asked. In a two-player public-good game with $u_i = \sqrt{(w - g_i)\, m\, (g_i + g_j)}$: find the best responses, the Nash equilibrium and the socially optimal contribution, then the discount factors for which grim trigger sustains the socially optimal contribution in the infinitely repeated game.

(b) Method. Best responses and the stage Nash equilibrium, then grim trigger with the one-stage deviation principle and normalised payoffs (Lecture 5, slides 21, 24–29; Tadelis §10.3, pp. 196–202). The incentive constraint compares the cooperative stage payoff with the best deviation followed by the stage Nash payoff forever.

(c) Solution. (a) Best response. Squaring does not change the maximiser, so player i maximises $(w - g_i)(g_i + g_j)$: $(w - g_i) - (g_i + g_j) = 0$, hence

$$g_i(g_j) = \frac{w - g_j}{2} = \frac{w}{2} - \frac{1}{2}\, g_j.$$

A rival's contribution crowds out one half of one's own: the response is $\frac{w}{2}$ when $g_j = 0$ and falls to 0 at $g_j = w$. The second-order condition holds, since $(w - g_i)(g_i + g_j)$ is concave in g_i.

(b) Nash equilibrium. Symmetric: $g = \frac{w-g}{2}$, so $g^* = \frac{w}{3}$. Payoff $u_i^* = \sqrt{\frac{2w}{3} \cdot m \cdot \frac{2w}{3}} = \frac{2w}{3}\sqrt{m}$.

(c) Social optimum. Maximise $u_A + u_B$. At a symmetric profile each term is $\sqrt{(w-g)\,2mg}$, so the joint problem is to maximise $(w-g)g$, with solution $g^{SO} = \frac{w}{2}$ (the first-order condition for the asymmetric problem is also satisfied at $g_A = g_B = \frac{w}{2}$). Payoff $u_i^{SO} = \sqrt{\frac{w}{2} \cdot m \cdot w} = \frac{w\sqrt{m}}{\sqrt{2}} \approx 0.707\, w\sqrt{m}$, against $u_i^* = 0.667\, w\sqrt{m}$: the difference is $\frac{3\sqrt{2}-4}{6}\, w\sqrt{m} \approx 0.04\, w\sqrt{m} > 0$. Each player is better off at the social optimum, which has a larger contribution ($\frac{w}{2} > \frac{w}{3}$): at Nash, each ignores the benefit his donation gives the other and free-rides.

(d) Grim trigger. Strategy: contribute $\frac{w}{2}$ as long as both have; otherwise contribute $\frac{w}{3}$ forever. Per-period payoffs, in units of $w\sqrt{m}$:

cooperation	$\frac{1}{\sqrt{2}}$	both give $\frac{w}{2}$
best deviation	$\frac{3}{4}$	$g_i = \frac{w - w/2}{2} = \frac{w}{4}$ against $g_j = \frac{w}{2}$: $\sqrt{\frac{3w}{4}\, m\, \frac{3w}{4}} = \frac{3w}{4}\sqrt{m}$
punishment (Nash)	$\frac{2}{3}$	both give $\frac{w}{3}$

The punishment state needs no check (a stage equilibrium). The cooperative state requires, in normalised payoffs,

$$\frac{1}{\sqrt{2}} \;\geq\; (1-\delta)\,\frac{3}{4} + \delta\,\frac{2}{3} \;\Longleftrightarrow\; \delta \;\geq\; \frac{\frac{3}{4} - \frac{1}{\sqrt{2}}}{\frac{3}{4} - \frac{2}{3}} = 12\Big(\frac{3}{4} - \frac{1}{\sqrt{2}}\Big) = 9 - 6\sqrt{2} \approx 0.5147.$$

This is (10.5), $\delta^* = (v^d - v^c)/(v^d - v^*)$. The threshold does not depend on w or m, because both scale every payoff by the same factor.

(d) Interpretation. Cooperation is worth only 0.04 more than Nash (a 6% gain) while a deviation pays 0.043 more than cooperating. The two are of the same size, so the players must be fairly patient, $\delta \geq 0.51$.

(e) Verification. `check_L05.py` reproduces the book's answers with sympy: $g_i(g_j) = \frac{w-g_j}{2}$, $g^* = \frac{w}{3}$, $g^{SO} = \frac{w}{2}$, payoffs $\frac{w\sqrt{m}}{\sqrt{2}}$, $\frac{2w\sqrt{m}}{3}$, deviation $\frac{3w\sqrt{m}}{4}$, and $\delta^* = 9 - 6\sqrt{2} = 0.5147$ (the book: "$\delta \geq 0.51$"); the inequality holds at 0.52 and fails at 0.51.

Answer. $g_i(g_j) = \frac{w-g_j}{2}$; $g^* = \frac{w}{3}$; $g^{SO} = \frac{w}{2}$ (higher utility for both: $\frac{w\sqrt{m}}{\sqrt{2}}$ against $\frac{2w\sqrt{m}}{3}$); grim trigger sustains g^{SO} iff $\delta \geq 9 - 6\sqrt{2} \approx 0.51$ (best deviation $g = \frac{w}{4}$ with payoff $\frac{3w\sqrt{m}}{4}$).

Item 15: Muñoz-Garcia Ch. 6 Exercise 9

Exam level · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 9 "Representing Feasible, Individually Rational Payoffs", p. 305 [PDF 316]. Answer in the source: yes, a worked answer follows the exercise. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider an infinitely-repeated game where the stage game is depicted Table 6.6. Player 1 chooses rows and player 2 chooses columns. Players discount the future using a common discount factor δ.
>
> **Part (a)** What outcomes in the stage-game are consistent with Nash equilibrium play?
>
> **Part (b)** Let v_1 and v_2 be the repeated game payoffs to player 1 and player 2 respectively. Draw the set of feasible payoffs from the repeated game, explaining any normalization you use.
>
> **Part (c)** Find the set of individually rational feasible set of payoffs.
>
> **Part (d)** Find a Nash equilibrium in which the players obtain (9, 9) each period. What restrictions on δ are necessary?

Player 1 Player 2	L	R
U	9, 9	1, 10
D	10, 1	7, 7

Table 6.6 Simultaneous-move game

(a) What is asked. For the infinitely repeated 2×2 game of Table 6.6: (a) the stage Nash equilibrium; (b) the feasible payoff set and the normalization used; (c) the feasible, individually rational set; (d) a Nash equilibrium that gives $(9, 9)$ every period, and the condition it needs on δ.

(b) Method. Stage equilibrium by strict dominance (Tadelis §4.1). Feasible payoffs as the convex hull of the stage payoffs (Tadelis Defs. 10.6–10.7, p. 209–210) with payoffs measured as average discounted payoffs (Def. 10.3, p. 194). Individual rationality relative to the minimax payoff, and the folk theorem (Theorem 10.1, p. 211; MWG Prop. 12.AA.5, p. 422, for the minimax version). Part (d) uses grim trigger and the threshold (10.5), p. 202, checked by the one-stage deviation principle (Prop. 10.3, p. 198). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-06-folk-theorem.md`.

(c) Solution. Notation. $q = \Pr(L)$ in a mixed strategy of player 2, $p = \Pr(U)$ for player 1. The average payoff of a stream is $\bar{v}_i = (1-\delta)\sum_{t\geq 1} \delta^{t-1} u_i(a^t)$. Best replies are underlined:

	L	R
U	9, 9	1, $\underline{10}$
D	$\underline{10}$, 1	$\underline{7}$, $\underline{7}$

(a) Stage equilibrium. For player 1, D beats U against both columns ($10 > 9$, $7 > 1$), so D is strictly dominant; by symmetry R is strictly dominant for player 2. The only outcome

consistent with Nash equilibrium play is (D, R), payoffs $(7, 7)$. There is no mixed equilibrium: for every q, $u_1(D, q) = 10q + 7(1 - q) = 7 + 3q$ exceeds $u_1(U, q) = 9q + 1(1 - q) = 1 + 8q$, because $(7 + 3q) - (1 + 8q) = 6 - 5q > 0$ on $[0, 1]$. The game is a Prisoner's Dilemma: (U, L) gives both $9 > 7$.

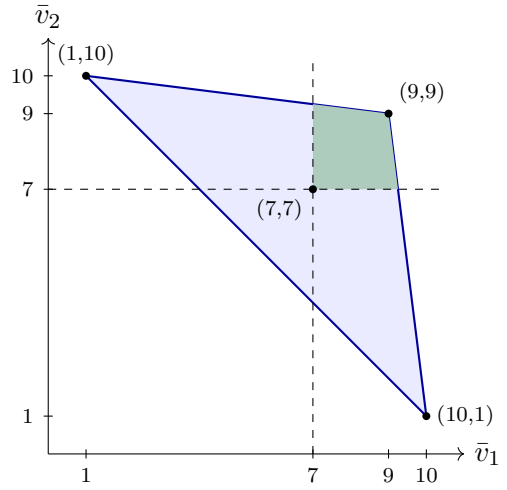
(b) Feasible payoffs. *Normalization.* Repeated-game payoffs are measured as average discounted payoffs, $\bar{v}_i = (1 - \delta)v_i$. With this scaling a constant stream x is worth x (Tadelis (10.2), p. 195), so repeated-game payoffs are in the same units as stage payoffs and can be drawn in the same plane. *Convexification.* The four pure payoff vectors are $(9, 9)$, $(1, 10)$, $(10, 1)$, $(7, 7)$. Every point of their convex hull is reached by playing the outcomes with the corresponding long-run frequencies (exactly with a public randomization device, as in the book, or approximately for δ near 1 with deterministic sequences). The feasible set is

$$V = \text{CoHull}\{(9, 9), (1, 10), (10, 1), (7, 7)\}.$$

Its shape. $(7, 7)$ lies strictly inside the triangle with vertices $(1, 10)$, $(10, 1)$, $(9, 9)$, since

$$(7, 7) = \tfrac{3}{7}(9, 9) + \tfrac{2}{7}(1, 10) + \tfrac{2}{7}(10, 1),$$

with all weights positive. So V is that triangle. Its edges are: $v_1 + v_2 = 11$ from $(1, 10)$ to $(10, 1)$; $v_2 = 10 - \frac{v_1 - 1}{8}$ from $(1, 10)$ to $(9, 9)$; and $v_1 = 10 - \frac{v_2 - 1}{8}$ from $(10, 1)$ to $(9, 9)$.



Feasible set V (triangle) and the FIR set (shaded, without its edges on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$).

(c) Individually rational payoffs. Player 1's minimax payoff is

$$\underline{v}_1 = \min_{q \in [0,1]} \max\{u_1(U, q),\, u_1(D, q)\} = \min_q\, (7 + 3q) = 7 \quad (q = 0,\ \text{i.e. } R),$$

because D is the best reply for every q. By symmetry $\underline{v}_2 = 7$ (player 1 plays D). Here the minimax payoff equals the stage Nash payoff. The feasible, individually rational (FIR) set is

$$\{\bar{v} \in V : \bar{v}_1 > 7,\ \bar{v}_2 > 7\}.$$

Cutting the triangle with $\bar{v}_1 = 7$ (on the edge $(1, 10)$–$(9, 9)$: $\bar{v}_2 = 10 - \frac{6}{8} = \frac{37}{4}$) and with $\bar{v}_2 = 7$ (on the edge $(10, 1)$–$(9, 9)$: $\bar{v}_1 = \frac{37}{4}$) gives the quadrilateral with corners

$$(7, 7), \quad (\tfrac{37}{4}, 7), \quad (9, 9), \quad (7, \tfrac{37}{4}),$$

without its two sides on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$. Its area is $\frac{9}{2}$.

(d) A Nash equilibrium with $(9, 9)$ every period. Grim trigger:

- player 1 plays U in period 1 and as long as every past outcome was (U, L); after any other history, D forever;
- player 2 plays L in period 1 and as long as every past outcome was (U, L); after any other history, R forever.

Cooperation phase. Staying gives $9 + 9\delta + 9\delta^2 + \cdots = \frac{9}{1-\delta}$. The best deviation is D (10 against L), followed by (D, R) forever: $10 + \frac{7\delta}{1-\delta}$. No deviation pays iff

$$\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta} \iff 9 \geq 10(1-\delta) + 7\delta = 10 - 3\delta \iff \delta \geq \frac{1}{3}.$$

(In average payoffs: $9 \geq (1-\delta)10 + \delta\, 7$.) By (10.5), $\delta^* = \frac{10-9}{10-7} = \frac{1}{3}$. Player 2's problem is identical by symmetry. *Punishment phase.* (D, R) is a stage equilibrium and the continuation does not depend on today's action, so deviating to U (or L) loses $7 - 1 = 6$ today and changes nothing later. Both checks pass, so the profile is a subgame-perfect equilibrium, and therefore a Nash equilibrium, for $\delta \geq \frac{1}{3}$.

The restriction is necessary. In any Nash equilibrium with path (U, L) forever, player 1 can deviate to D in period 1 and keep playing D. Since $u_1(D, \cdot) \geq 7$, this gives at least $10 + \frac{7\delta}{1-\delta}$ whatever player 2 does. So $\delta \geq \frac{1}{3}$ is needed for every Nash equilibrium, not just for grim trigger: the punishment already pushes the deviator down to his minimax payoff of 7.

(d) Interpretation. $(9, 9)$ is feasible and gives each player more than the minimax 7, so the folk theorem says it can be sustained by patient enough players. The exercise puts a number on "patient enough": the one-period gain from defecting is 1, the per-period loss from punishment is $9 - 7 = 2$, so $\delta^* = 1/(1+2) = \frac{1}{3}$. The FIR quadrilateral also contains asymmetric outcomes, e.g. near $(\frac{37}{4}, 7)$, which favour player 1 heavily. The folk theorem says these are sustainable too, which is why it explains how cooperation is possible rather than predicting which outcome occurs.

(e) Verification. `check_C_L05.py` confirms that (D, R) is the unique stage equilibrium and that $u_1(D, q) > u_1(U, q)$ for every q on a grid of step $\frac{1}{1000}$, with $\min_q \max = 7$. With sympy it computes the convex hull of the four payoff vectors (the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$, with $(7, 7)$ strictly inside), the corners $(7, \frac{37}{4})$ and $(\frac{37}{4}, 7)$ and the area $\frac{9}{2}$. It solves $\frac{9}{1-\delta} = 10 + \frac{7\delta}{1-\delta}$ to get $\delta = \frac{1}{3}$ and tests the inequality on the grid $k/1000$ and at $\frac{1}{3} \pm \frac{1}{1000}$. Output: "Munoz-Garcia 6.9: stage NE (D,R) (7,7) = minimax; hull = triangle (1,10),(10,1),(9,9); FIR corners (7,7),(37/4,7),(9,9),(7,37/4); delta* = 1/3 (book: 1/3)". *Comparison with Muñoz-Garcia's answer* (pp. 305–309): the same stage equilibrium, minimax $(7, 7)$, FIR set and threshold $\delta \geq \frac{1}{3}$, and the book also shows the strategies are subgame perfect. Three slips in the book: (i) it calls the feasible set "not convex" (p. 307). The convex hull is convex by definition; what is not convex is the set of the four pure payoff vectors, which is what the public randomization device fills in. (ii) It writes $u_2(L; p) = 9p + 1(1-p) = 7 + 3p$ (p. 306); the sum is $1 + 8p$, and $7 + 3p$ is $u_2(R; p)$. (iii) It refers to "Fig. 6.7" for the figure it has just drawn (Fig. 6.11). The book takes the public randomization device as the normalization of part (b); the average-payoff scaling above is the normalization usually meant, and both are needed to draw V in stage-payoff units.

(f) Common mistakes.

- Drawing the four points, or the quadrilateral with $(7, 7)$ as a vertex, as the feasible set. $(7, 7)$ is inside the triangle, so it is not a vertex of V; it is, however, a corner of the FIR set.
- Using ≥ 7 instead of > 7 for individual rationality, or using pure-strategy minimax without checking mixed strategies. Here both give 7, but the definition uses mixed strategies.

- Mixing present values and averages in one inequality, e.g. $9 \geq 10 + \frac{7\delta}{1-\delta}$. Both sides must use the same scaling.

Answer. (a) D and R are strictly dominant: the only stage equilibrium is (D, R), payoffs $(7, 7)$. (b) With average payoffs $(1-\delta)\sum \delta^{t-1} u_i$ and public randomization, V is the convex hull of the four payoff vectors, the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$; $(7, 7)$ is inside it. (c) Minimax $= (7, 7)$; FIR set $= \{\bar{v} \in V : \bar{v}_1 > 7, \bar{v}_2 > 7\}$, the quadrilateral $(7, 7)$, $(\frac{37}{4}, 7)$, $(9, 9)$, $(7, \frac{37}{4})$. (d) Grim trigger (play (U, L) until any deviation, then (D, R) forever) is an SPE, hence a Nash equilibrium, iff $\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta}$, i.e. $\delta \geq \frac{1}{3}$ (book: $\frac{1}{3}$). The same bound is necessary for any Nash equilibrium.

Item 16: Gibbons Problem 2.11

Exam level · difficulty 4/5

Statement (verbatim from the list, item 16)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §2.3 Problems, Problem 2.11, p. 135 [PDF 142]. Answer in the source: none in the book (Muñoz-Garcia Ch. 6 Ex. 7, p. 299, works the same game). Reproduced verbatim. (The same game is Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 7, p. 299, with a worked answer.)

> 2.11. The simultaneous-move game (below) is played twice, with the outcome of the first stage observed before the second stage begins. There is no discounting. Can the payoff $(4, 4)$ be achieved in the first stage in a pure-strategy subgame-perfect Nash equilibrium? If so, give strategies that do so. If not, prove why not.

	L	C	R
T	3, 1	0, 0	5, 0
M	2, 1	1, 2	3, 1
B	1, 2	0, 1	4, 4

(a) What is asked. For the 3×3 game played twice without discounting, decide whether $(4, 4)$ can be obtained in the first stage of a pure-strategy SPE, and give the strategies or a proof.

(b) Method. Two-stage repeated game with several stage equilibria (Gibbons §2.3.A, pp. 82–88, especially the three-strategy example of pp. 84–87; Tadelis §9.4, pp. 180–184, and §10.1, p. 191). In the second stage a stage Nash equilibrium must be played after every history (Tadelis Prop. 9.2, p. 181). A first-stage profile that is not a stage equilibrium is supported by a *reward* equilibrium after it and a *punishment* equilibrium after deviations. Deviations are checked one stage at a time (Tadelis Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, each period starts a subgame) and slide 17 (SPE). Notation as in `Notas/L05/L05-02-two-stage-carrot-stick.md`.

(c) Solution. **Notation.** $v_i = u_i(a^1) + u_i(a^2)$ (no discounting). Best replies are underlined:

	L	C	R
T	$\underline{3}$, $\underline{1}$	0, 0	$\underline{5}$, 0
M	2, 1	$\underline{1}$, $\underline{2}$	3, 1
B	1, 2	0, 1	4, $\underline{4}$

Step 1: stage equilibria. Player 1's best replies: to L, T (3); to C, M (1); to R, T (5). Player 2's: to T, L (1); to M, C (2); to B, R (4). The mutual best replies are

$$(T, L) \text{ with } (3, 1), \qquad (M, C) \text{ with } (1, 2).$$

(B, R) is *not* a stage equilibrium: against R, player 1 prefers T ($5 > 4$). Since the question asks for a pure-strategy SPE, second-stage play must be one of these two pure equilibria after every history.

Step 2: who must be deterred. At (B, R), player 2 is already best-responding (4 is the most she can get against B), so only player 1 has a profitable one-stage deviation: to T, gaining

$5 - 4 = 1$ (to M he would get $3 < 4$). Player 1 gets 3 in (T, L) and 1 in (M, C), so the second stage can reward him with 3 and punish him with 1: a swing of $2 > 1$.

Step 3: the strategies.

- Player 1: play B in stage 1; in stage 2 play T if the stage-1 outcome was (B, R), and M otherwise.
- Player 2: play R in stage 1; in stage 2 play L if the stage-1 outcome was (B, R), and C otherwise.

Step 4: one-stage deviation checks. *Stage 2 (every history).* The prescribed profile is (T, L) or (M, C), a stage equilibrium, and nothing follows, so no deviation pays.

Stage 1, player 1. On the path he gets $4 + 3 = 7$. Deviating to T gives 5 now and then (M, C), so $5 + 1 = 6 < 7$; deviating to M gives $3 + 1 = 4 < 7$.

Stage 1, player 2. On the path she gets $4 + 1 = 5$. Deviating to L gives 2 now and then (M, C), so $2 + 2 = 4 < 5$; deviating to C gives $1 + 2 = 3 < 5$. (The "punishment" (M, C) is actually better for player 2 than (T, L), but she has nothing to gain in stage 1, so it does not matter.)

All one-stage deviations lose, so the profile is a pure-strategy SPE and the answer is **yes**: $(4, 4)$ is reached in the first stage, with total payoffs $(7, 5)$.

How much room there is. With a discount factor δ, player 1's constraint is $4 + 3\delta \geq 5 + \delta \iff \delta \geq \frac{1}{2}$, so the construction survives some impatience; player 2's constraint, $4 + \delta \geq 2 + 2\delta \iff \delta \leq 2$, holds for every $\delta \in [0, 1]$. Among all $2^9 = 512$ assignments of (T, L) or (M, C) to the nine stage-1 outcomes, exactly $128 = 2^7$ support (B, R) first: those with (T, L) after (B, R) and (M, C) after player 1's deviation to T. The other seven outcomes can be assigned either way.

(d) Interpretation. The second stage offers two equilibria that player 1 ranks differently, and that difference is a credible incentive. The players can coordinate on the efficient (B, R) today because both expect player 1 to be "paid" with his preferred equilibrium (T, L) tomorrow if he behaves, and pushed to (M, C) if he grabs the extra 1. Gibbons (pp. 87–88) warns that such threats may be renegotiated when the punishment equilibrium is Pareto-dominated by the reward equilibrium. That is not the case here: player 2 prefers (M, C) to (T, L), so after player 1's deviation she would not agree to forgive him, and the threat survives that critique.

(e) Verification. `check_C_L05.py` finds the two pure stage equilibria, confirms that player 1's best reply to R is T, and enumerates all 512 continuation maps with (B, R) first at $\delta = 1$. It finds 128 SPE, all with (T, L) after (B, R) and (M, C) after (T, R). It checks the payoffs 7 against 6, and on the grid $\delta = k/1000$ that the constructed profile works exactly for $\delta \geq \frac{1}{2}$. Output: "Gibbons 2.11: stage NE (T,L),(M,C); (B,R) first is a pure SPE: 128 of 512 maps work, all with (T,L) after (B,R) and (M,C) after (T,R)". Gibbons prints no answer. *Comparison with Muñoz-Garcia Ch. 6 Ex. 7* (pp. 299–303, the same game): the same answer (yes, (B, R) then (T, L), with $7 > 6$ for player 1). Two differences. (i) The book punishes with the stage game's mixed equilibrium (each player mixes $\frac{1}{2}$–$\frac{1}{2}$ over $\{T, M\}$ and $\{L, C\}$, payoffs $(\frac{3}{2}, 1)$), giving $5 + \frac{3}{2} = 6.5 < 7$. That works as an SPE, but it is not a *pure-strategy* SPE as the question asks; the pure equilibrium (M, C) is both admissible and a harsher punishment. (ii) The book lists $BR_1(R) = B$; in fact $BR_1(R) = T$ $(5 > 4)$, which is exactly why (B, R) needs support. Its list of pure equilibria, (T, L) and (M, C), is right.

(f) Common mistakes.

- Answering "no, because (B, R) is not a Nash equilibrium of the stage game". In a repeated game with several stage equilibria, non-equilibrium play *can* be supported in early stages.

- Using a mixed equilibrium as the punishment when the question restricts to pure strategies, or a non-equilibrium profile (e.g. (B, C)) in the last stage.
- Checking only player 2 (whose payoff 4 is high) and missing that player 1 is the one tempted.

Answer. Yes. Stage equilibria: (T, L) $(3, 1)$ and (M, C) $(1, 2)$. Strategies: play (B, R) in stage 1; in stage 2 play (T, L) if stage 1 was (B, R) and (M, C) otherwise. Player 1: $4 + 3 = 7 > 5 + 1 = 6$ (deviation to T) and $> 3 + 1$ (to M); player 2: $4 + 1 = 5 > 2 + 2$ (to L) and $> 1 + 2$ (to C); stage 2 is always a stage equilibrium. This is a pure-strategy SPE with $(4, 4)$ in the first stage (it would survive discounting down to $\delta = \frac{1}{2}$).

Item 17: Tadelis Exercise 9.1

Exam level · difficulty 4/5

Statement (verbatim from the list, item 17)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.1 "Multistage Practice", pp. 186–187 [PDF 205–206]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Multistage Practice:** Consider the following simultaneous-move game that is played twice (the players observe the first-period outcome prior to the second-period play):

Player 1 Player 2	L	C	R
T	10, 10	2, 12	0, 13
M	12, 2	5, 5	0, 0
B	13, 0	0, 0	1, 1

> a. Find all the pure-strategy subgame-perfect equilibria with no discounting ($\delta = 1$). Be precise in defining history-contingent strategies for both players.
>
> b. For each of the equilibria you found in (a), find the smallest discount factor that supports it.

(a) What is asked. For the 3×3 game played twice with the first-period outcome observed: (a) all pure-strategy SPE with $\delta = 1$, with history-contingent strategies; (b) for each, the smallest δ that supports it.

(b) Method. Multistage games (Tadelis §9.1–9.4, pp. 175–184) and the finitely repeated game $G(2, \delta)$ (§10.1, pp. 190–192). Period-2 play must be a stage Nash equilibrium after every history (Prop. 9.2, p. 181). A first-period profile that is not a stage equilibrium can be supported only by a carrot (a good stage equilibrium on the path) and a stick (a bad one after a deviation), and only when $\delta \geq \delta^* = g/(c - s)$ (`Notas/L04/L04-02`, `Notas/L05/L05-02`; Tadelis's own two-stage example, p. 191). Deviations are checked one stage at a time (one-stage deviation principle, Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, payoff $\sum_t \delta^{t-1} u_i^t$, each period starts a subgame) and slide 17 (SPE).

(c) Solution. Notation. $v_i = u_i(a^1) + \delta\, u_i(a^2)$, where a^t is the period-t action profile. A pure strategy for player i is a period-1 action and a map from the nine possible period-1 outcomes to a period-2 action. $g_i(a)$ is player i's best one-shot gain from deviating from a:

$$g_1(a) = \max_r u_1(r, a_2) - u_1(a), \qquad g_2(a) = \max_c u_2(a_1, c) - u_2(a).$$

Best replies are underlined:

	L	C	R
T	10, 10	2, 12	0, $\underline{13}$
M	12, 2	$\underline{5}$, $\underline{5}$	0, 0
B	$\underline{13}$, 0	0, 0	$\underline{1}$, $\underline{1}$

Step 1: stage equilibria. Player 1's best replies: to L, B (13); to C, M (5); to R, B (1). Player 2's: to T, R (13); to M, C (5); to B, R (1). The mutual best replies are

$$(M, C) \text{ with } (5, 5) \quad \text{(the carrot)}, \qquad (B, R) \text{ with } (1, 1) \quad \text{(the stick)}.$$

Each player gets 5 in the carrot and 1 in the stick, so the largest punishment one period can deliver is $c - s = 5 - 1 = 4$.

Step 2: period 2 (the last phase). After any first-period outcome the subgame is the stage game (plus a sunk constant), so it must be (M, C) or (B, R). No one-stage deviation in period 2 pays, because both are stage equilibria and nothing follows.

Step 3: period 1 (which profiles can be supported). A first-period profile a is supportable if and only if each player's gain is at most what the threat can take away:

$$g_i(a) \;\le\; \delta\,(5 - 1) = 4\delta, \qquad i = 1, 2, \tag{1}$$

using (M, C) after a and (B, R) after any deviation. No continuation can do better, since 5 is the most and 1 the least a player can get in period 2. For a stage equilibrium a, $g_i(a) = 0$ and any continuation works. The gains are:

a	$u(a)$	$g_1(a)$	$g_2(a)$	supportable ($\delta = 1$)?	$\delta^* = \max_i g_i/4$
(T, L)	$(10, 10)$	$13 - 10 = 3$	$13 - 10 = 3$	yes	$3/4$
(T, C)	$(2, 12)$	$5 - 2 = 3$	$13 - 12 = 1$	yes	$3/4$
(T, R)	$(0, 13)$	$1 - 0 = 1$	0	yes	$1/4$
(M, L)	$(12, 2)$	$13 - 12 = 1$	$5 - 2 = 3$	yes	$3/4$
(M, C)	$(5, 5)$	0	0	yes	0
(M, R)	$(0, 0)$	$1 - 0 = 1$	$5 - 0 = 5$	no	–
(B, L)	$(13, 0)$	0	$1 - 0 = 1$	yes	$1/4$
(B, C)	$(0, 0)$	$5 - 0 = 5$	$1 - 0 = 1$	no	–
(B, R)	$(1, 1)$	0	0	yes	0

(a) The pure SPE with $\delta = 1$. There are nine SPE *paths*:

- the four sequences of stage equilibria, played regardless of history (Prop. 9.1, p. 180): (M, C) then (M, C); (M, C) then (B, R); (B, R) then (M, C); (B, R) then (B, R);
- the five carrot-and-stick paths: a then (M, C), for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$.

For a carrot-and-stick path the strategies are, for $\hat{a} = (\hat{a}_1, \hat{a}_2)$ one of those five:

$$s_1 : \text{ play } \hat{a}_1; \text{ in period 2 play } M \text{ if } \hat{a} \text{ was played, } B \text{ otherwise;}$$
$$s_2 : \text{ play } \hat{a}_2; \text{ in period 2 play } C \text{ if } \hat{a} \text{ was played, } R \text{ otherwise.}$$

One-stage deviation checks for each path (with $\delta = 1$; path payoff against best deviation):

$\hat{a}$	P1: path	P1: best deviation	P2: path	P2: best deviation
(T, L)	$10 + 5 = 15$	B: $13 + 1 = 14$	$10 + 5 = 15$	R: $13 + 1 = 14$
(T, C)	$2 + 5 = 7$	M: $5 + 1 = 6$	$12 + 5 = 17$	R: $13 + 1 = 14$
(T, R)	$0 + 5 = 5$	B: $1 + 1 = 2$	$13 + 5 = 18$	none (best reply)
(M, L)	$12 + 5 = 17$	B: $13 + 1 = 14$	$2 + 5 = 7$	C: $5 + 1 = 6$
(B, L)	$13 + 5 = 18$	none (best reply)	$0 + 5 = 5$	R: $1 + 1 = 2$

In period 2 both players play a stage equilibrium after every history, so no deviation pays there. (M, R) and (B, C) cannot be supported: a gain of 5 exceeds the largest punishment, 4. For instance, at (M, R) player 2 gets $0 + 5 = 5$ on the path but $5 + 1 = 6$ by deviating to C.

These strategies are not the only ones. Any assignment of (M, C) or (B, R) to the nine period-1 outcomes that satisfies (1) for the deviations that would be profitable also works (e.g. for (T, R) only player 1's deviation to B must be punished). Counting every such map, `check_C_L05.py` finds 992 pure SPE strategy profiles at $\delta = 1$, all with one of the nine paths above.

(b) The smallest discount factor. From (1), $\delta^*(a) = \max_i g_i(a)/4$ (last column of the first table). Solving each binding inequality:

- (T, L): $10 + 5\delta \geq 13 + \delta \iff 4\delta \geq 3 \iff \delta \geq \frac{3}{4}$ (both players);
- (T, C): player 1 binds, $2+5\delta \geq 5+\delta \iff \delta \geq \frac{3}{4}$; player 2 needs $12+5\delta \geq 13+\delta \iff \delta \geq \frac{1}{4}$;
- (M, L): the mirror image of (T, C), $\delta \geq \frac{3}{4}$;
- (T, R): only player 1, $0 + 5\delta \geq 1 + \delta \iff \delta \geq \frac{1}{4}$;
- (B, L): only player 2, $0 + 5\delta \geq 1 + \delta \iff \delta \geq \frac{1}{4}$;
- the four sequences of stage equilibria: any $\delta \geq 0$.

Below δ^* no continuation map works, because (M, C) after $\hat{a}$ and (B, R) after the deviation is already the largest reward gap available.

A caveat: renegotiation. The stick (B, R) is Pareto-dominated by (M, C). After a deviation both players would like to "forgive" and play (M, C) in period 2, and if they can renegotiate the threat fails (Gibbons §2.3.A, pp. 87–88; `Notas/L05/L05-02`). The SPE above take the threat as given.

(d) Interpretation. Two stage equilibria with different payoffs give the second period a carrot and a stick. That lets the players reach (T, L), which pays $(10, 10)$, twice the $(5, 5)$ of the good stage equilibrium, in the first period. Profiles with a gain above 4 cannot be enforced by a single future period, however patient the players are. This is the finite-horizon version of the logic of Lecture 5: what can be sustained today is limited by how much the future can reward and punish. With a unique stage equilibrium (item Cabral 7.13) that gap is zero and nothing can be sustained.

(e) Verification. `check_C_L05.py` finds the two stage equilibria and enumerates all 9×2^9 combinations of first-period profile and continuation map at $\delta = 1$ (992 SPE, 9 paths; (M, R) and (B, C) excluded). For every threshold it checks that each path is an SPE at δ^* and at $\delta^* + \frac{1}{1000}$, that no continuation map at all supports it at $\delta^* - \frac{1}{1000}$, and that on the grid $\delta = k/1000$ the carrot-and-stick map works exactly when $\delta \geq \delta^*$. Output: "Tadelis 9.1: stage NE (M,C),(B,R); 992 pure SPE at delta=1 on 9 paths; delta* = 3/4 (TL, TC, ML), 1/4 (TR, BL); grid and +-1/1000 pass". Tadelis prints no answer.

(f) Common mistakes.

- Listing only the outcome "(T, L) then (M, C)" without the off-path play. The question asks for history-contingent strategies, and the threat to play (B, R) after a deviation is the whole point.
- Using the smallest gain instead of the largest: the threshold is $\max_i g_i/4$, so in (T, C) player 1's gain of 3 binds, not player 2's gain of 1.
- Claiming (M, R) or (B, C) can be sustained with enough patience. With one period left the punishment is capped at $5 - 1 = 4$, which is below the gain of 5 even at $\delta = 1$.

Answer. Stage equilibria: (M, C) $(5, 5)$ and (B, R) $(1, 1)$. (a) Pure SPE paths with $\delta = 1$: the four sequences of stage equilibria, and a then (M, C) for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$, supported by "play (M, C) after a, (B, R) after any other outcome". (M, R) and (B, C) cannot be supported. (b) $\delta^* = \frac{3}{4}$ for (T, L), (T, C), (M, L); $\frac{1}{4}$ for (T, R), (B, L); 0 for the sequences of stage equilibria.

Item 18: Tadelis Exercise 10.5

Exam level · difficulty 4/5

Statement (verbatim from the list, item 18)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.5 "Negative Ad Campaigns Revisited", pp. 216–217 [PDF 235–236]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Negative Ad Campaigns Revisited:** Recall exercise 5.14 from Chapter 5, in which each one of two political parties can choose to buy time on commercial radio shows to broadcast negative ad campaigns against its rival. These choices are made simultaneously. Government regulations forbid a party from buying more than 2 hours of negative campaign time, so that each party cannot choose an amount of negative campaigning above 2 hours. Given a pair of choices (a_1, a_2), the payoff of party i is given by the following function: $v_i(a_1, a_2) = a_i - 2a_j + a_i a_j - (a_i)^2$.
>
> a. Find the unique pure-strategy Nash equilibrium of the one-shot game.
>
> b. If the parties could sign a binding agreement on how much to campaign, what levels would they choose?
>
> c. Now assume that this game is repeated infinitely often, and the foregoing demonstrates the choices and payoffs per period. For which discount factors $\delta \in (0, 1)$ can the levels you found in (b) be supported as a subgame-perfect equilibrium of the infinitely repeated game?
>
> d. Despite the parties' ability to coordinate, as you have demonstrated in your answer to (c), the government is concerned about the parties' ability to run up to 2 hours a day of negative campaigning, and it is considering limiting negative campaigning to $\frac{1}{2}$ hour a day, so that now $a_i \in [0, \frac{1}{2}]$. Is this a good policy to limit negative campaigns further? Justify your answer with the relevant calculations. What is the intuition for your conclusion?

(a) What is asked. With $v_i = a_i - 2a_j + a_i a_j - a_i^2$ and $a_i \in [0, 2]$: find the stage equilibrium and the binding-agreement optimum; the discount factors that sustain the optimum by grim trigger; and whether capping campaigns at $\frac{1}{2}$ hour is a good policy.

(b) Method. Grim trigger with continuous actions and normalised payoffs (Lecture 5, slides 21, 24–29; Tadelis §10.3, pp. 196–202, threshold (10.5), p. 202). For part (d) the threshold is recomputed on the new action set, and the policy is judged by the level of campaigning each regime sustains.

(c) Solution. **(a) Stage equilibrium.** $\partial v_i / \partial a_i = 1 + a_j - 2a_i = 0$ gives the best response $a_i = \frac{1+a_j}{2}$ (always inside $[0, 2]$ for $a_j \in [0, 2]$), and v_i is concave in a_i. The unique equilibrium solves $a = \frac{1+a}{2}$: $a_1 = a_2 = 1$, with $v_i = 1 - 2 + 1 - 1 = -1$.

(b) Binding agreement. Joint payoff:

$$v_1 + v_2 = -(a_1 + a_2) - (a_1 - a_2)^2 \;\leq\; 0,$$

with equality only at $a_1 = a_2 = 0$. They choose $(0, 0)$ with $v_i = 0 > -1$.

(c) Grim trigger for $(0, 0)$. Strategy: campaign 0 as long as both have; otherwise campaign 1 forever. Punishment state: a stage equilibrium, so no check. Cooperative state: against $a_j = 0$

the best response is $a_i = \frac{1}{2}$, giving $v^d = \frac{1}{2} - 0 + 0 - \frac{1}{4} = \frac{1}{4}$. In normalised payoffs

$$0 \;\geq\; (1-\delta)\cdot\tfrac{1}{4} + \delta\cdot(-1) \;\iff\; \tfrac{1}{4} \;\geq\; \tfrac{5}{4}\,\delta \;\iff\; \delta \;\geq\; \frac{1}{5}$$

(check with (10.5): $\frac{1/4-0}{1/4+1} = \frac{1}{5}$). By symmetry the same holds for party 2. So $(0,0)$ is an SPE outcome iff $\delta \in [\frac{1}{5}, 1)$.

(d) The cap $a_i \in [0, \frac{1}{2}]$. The best response $\frac{1+a_j}{2} \geq \frac{1}{2}$ for every $a_j \geq 0$, so the constrained best response is always $\frac{1}{2}$: the new stage equilibrium is $(\frac{1}{2}, \frac{1}{2})$ with $v_i = \frac{1}{2} - 1 + \frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$. The cooperative outcome is still $(0,0)$ with $v = 0$ and the best deviation is still $\frac{1}{2}$ with $v^d = \frac{1}{4}$ (it is allowed). Only the punishment has changed:

$$0 \geq (1-\delta)\tfrac{1}{4} + \delta(-\tfrac{1}{2}) \;\iff\; \delta \;\geq\; \frac{1/4}{1/4+1/2} = \frac{1}{3}.$$

Is the cap a good policy? Compare the level of negative campaigning that each regime sustains. On the path of a symmetric SPE with campaigning a each, $v(a,a) = -a$; the best deviation and the punishment give, with the unrestricted action set (stage equilibrium -1),

$$-a \;\geq\; (1-\delta)\Big[\tfrac{(1+a)^2}{4} - 2a\Big] - \delta \;\iff\; a \;\geq\; a_{\min}(\delta) = \max\Big\{0,\; \frac{1-5\delta}{1-\delta}\Big\},$$

and with the cap (stage equilibrium $-\frac{1}{2}$, best deviation $\frac{1}{2}$ with payoff $\frac{1}{4} - \frac{3}{2}a$)

$$-a \;\geq\; (1-\delta)(\tfrac{1}{4} - \tfrac{3}{2}a) - \tfrac{\delta}{2} \;\iff\; (\tfrac{1}{2} - \tfrac{3}{2}\delta)(a - \tfrac{1}{2}) \;\geq\; 0,$$

so with the cap only $a = \frac{1}{2}$ is sustainable if $\delta < \frac{1}{3}$, and any $a \in [0, \frac{1}{2}]$ (so $a = 0$) if $\delta \geq \frac{1}{3}$. The lowest sustainable campaigning level is therefore:

δ	$\frac{1}{20}$	$\frac{1}{9}$	$\frac{1}{5}$	$\frac{1}{3}$ and above
no cap, $a_{\min}$	$\frac{15}{19} \approx 0.79$	$\frac{1}{2}$	0	0
cap at $\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0

The cap lowers negative campaigning only for $\delta < \frac{1}{9}$, where the parties are too impatient to collude and the cap simply stops them overshooting. For $\delta \in (\frac{1}{9}, \frac{1}{3})$ it *raises* campaigning to $\frac{1}{2}$ hour a day (from less than $\frac{1}{2}$, and for $\delta \geq \frac{1}{5}$ from 0), and for $\delta \geq \frac{1}{3}$ it changes nothing. Where the parties can coordinate without the cap ($\delta \geq \frac{1}{5}$, part (c)), the cap is either harmful ($\delta < \frac{1}{3}$) or irrelevant, so it is not a good policy.

(d) Interpretation. The cap lowers the cost of being punished: a milder stage equilibrium ($-\frac{1}{2}$ instead of -1) is a weaker threat, so collusion on *no* negative campaigning needs more patience ($\frac{1}{3}$ against $\frac{1}{5}$). Limiting actions helps in a one-shot game but, in a repeated game, it removes the stick that supports cooperation.

(e) Verification. `check_L05.py` solves the equilibrium $(1,1)$ with $v = -1$, shows $v_1 + v_2 = -(a_1 + a_2) - (a_1 - a_2)^2$, the deviation $\frac{1}{4}$, $\delta^* = \frac{1}{5}$ (both sides tested), the capped case $v = -\frac{1}{2}$ and $\delta^* = \frac{1}{3}$, verifies that $a = 1$ is a root of the unrestricted constraint and the other root is $\frac{1-5\delta}{1-\delta}$, the capped factorisation, and the table values.

(f) Common mistakes.

- Keeping the punishment payoff -1 in (d): with the cap the Nash payoff is $-\frac{1}{2}$.
- Judging the cap by the one-shot equilibrium alone ($1 \to \frac{1}{2}$): that is the right comparison only when $\delta < \frac{1}{9}$.

Answer. (a) $a_1 = a_2 = 1$, $v_i = -1$. (b) $(0,0)$, $v_i = 0$. (c) $\delta \geq \frac{1}{5}$ (best deviation $\frac{1}{2}$, payoff $\frac{1}{4}$). (d) With the cap the stage equilibrium is $(\frac{1}{2}, \frac{1}{2})$ with $v = -\frac{1}{2}$ and $(0,0)$ needs $\delta \geq \frac{1}{3}$. The cap helps only if $\delta < \frac{1}{9}$; for $\delta \in (\frac{1}{9}, \frac{1}{3})$ it makes campaigning higher ($\frac{1}{2}$ against less than $\frac{1}{2}$, or 0), and for $\delta \geq \frac{1}{3}$ it changes nothing. So it is not a good policy for patient parties.

Item 19: Tadelis Exercise 9.6

Exam level · difficulty 4/5

Statement (verbatim from the list, item 19)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.6 "Augmented Competition", p. 189 [PDF 208]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Augmented Competition:** Consider two firms playing a two-stage game with discount factor δ. In the first stage they play a Cournot quantity-setting game in which each firm has costs $c_i(q_i) = 10q_i$ for $i \in \{1, 2\}$ and the demand is given by $p(q) = 100 - q$, where $q = q_1 + q_2$. In the second stage, after the results of the Cournot game are observed, the firms play the following standard-setting game:
>
> | Player 1 Player 2 | a | b |
> |---|---|---|
> | A | 100, 100 | 0, 0 |
> | B | 0, 0 | 300, 300 |
>
> a. Find the unique Nash equilibrium of the first-stage game and the two pure-strategy Nash equilibria of the second-stage game.
>
> b. As far as the two firms are considered, what are the symmetric Pareto-optimal outcomes of each stage-game?
>
> c. For which values of δ can the Pareto-optimal outcomes be supported as a subgame-perfect equilibrium?
>
> d. Assume that $\delta = 0.5$. What is the "best" symmetric subgame-perfect equilibrium that the players can support?
>
> e. What happens to the best symmetric subgame-perfect equilibrium that the players can support as δ drops toward zero?

(a) What is asked. A Cournot game ($c = 10$, $p = 100 - q$) followed by a standard-setting game: find the stage equilibria and the symmetric Pareto-optimal outcomes, the δ that support them, the best symmetric SPE at $\delta = 0.5$, and what happens as $\delta \to 0$.

(b) Method. Carrot and stick over a continuous first stage (Tadelis §9.4, pp. 180–184; Thm. 9.1, p. 185). Stage 2 supplies the carrot (B, b) and the stick (A, a); the stage-1 constraint is *(stage-1 gain from deviating)* $\leq \delta$ [carrot − stick] (Lecture 5, slide 15). The Cournot model is the one of the slides' homework (slide 34).

(c) Solution. **(a) Stage equilibria.** *Stage 1:* $\pi_i = (90 - q_i - q_j)q_i$, best response $q_i = \frac{90 - q_j}{2}$, equilibrium $q_1 = q_2 = 30$, price 40, profit 900 each. *Stage 2:*

	a	b
A	<u>100</u>, <u>100</u>	0, 0
B	0, 0	<u>300</u>, <u>300</u>

The two pure equilibria are (A, a) with $(100, 100)$ and (B, b) with $(300, 300)$.

(b) Pareto-optimal symmetric outcomes. *Stage 1:* joint profit $(90-Q)Q$ is maximal at $Q=45$ (price 55, profit 2025), so $q_i=22.5$ and $\pi_i=1012.5>900$. *Stage 2:* (B,b) with $(300,300)$.

(c) Support. Strategies: $q_i=22.5$ in stage 1; in stage 2 play (B,b) after $(22.5,22.5)$ and (A,a) after any other history. Stage 2 is a stage equilibrium after every history, so only stage 1 needs checking. The best deviation against 22.5 is $q_i=\frac{90-22.5}{2}=33.75$, with profit $33.75^2=1139.0625$, followed by (A,a) worth 100:

$$1012.5+300\,\delta \;\geq\; 1139.0625+100\,\delta \iff 200\,\delta\geq 126.5625 \iff \delta \;\geq\; \frac{81}{128}\approx 0.633.$$

For $\delta\geq\frac{81}{128}$ the Pareto-optimal outcomes of both stages are an SPE path.

(d) $\delta=0.5<\frac{81}{128}$. The monopoly split is not supportable. The best symmetric SPE keeps (B,b) in stage 2 after the path and (A,a) after deviations, and chooses the lowest symmetric q whose deviation is deterred. A symmetric q earns $\pi(q)=(90-2q)q$ and the best deviation earns $\left(\frac{90-q}{2}\right)^2$, so the constraint is

$$\left(\tfrac{90-q}{2}\right)^2-(90-2q)q \;\leq\; \delta\,(300-100)=100 \iff \tfrac{9}{4}\,q^2-135\,q+2025\leq 100,$$

whose solutions are $q\in[\frac{70}{3},\,\frac{110}{3}]$ (roots $\frac{135\pm30}{4.5}$). Profit falls as q rises above 22.5, so the best symmetric SPE has

$$q_1=q_2=\tfrac{70}{3}\approx 23.33, \qquad \pi_i=\tfrac{130}{3}\cdot\tfrac{70}{3}=\tfrac{9100}{9}\approx 1011.1,$$

followed by (B,b): total value $1011.1+0.5\cdot 300\approx 1161.1$. At this q the incentive constraint holds with equality.

(e) As $\delta\to 0$. The same constraint with δ in place of $\frac{1}{2}$ reads $\frac{9}{4}q^2-135q+2025\leq 200\delta$, with lower root

$$q_{\min}(\delta)=30-\frac{20\sqrt{2}}{3}\sqrt{\delta}.$$

It equals $\frac{70}{3}$ at $\delta=\frac{1}{2}$, 22.5 at $\delta=\frac{81}{128}$, and tends to the Cournot quantity 30 as $\delta\to 0$. So the first-stage profit of the best symmetric SPE falls continuously to 900, while stage 2 still reaches (B,b) (the efficient stage equilibrium needs no future: it is a Nash equilibrium of the last stage on its own).

(d) Interpretation. The standard-setting game is a carrot for the Cournot game. With patience the firms restrain output to earn (B,b) later; with none, they play the Cournot equilibrium in stage 1 and still coordinate on the better standard in stage 2, because that is an equilibrium whatever happens.

(e) Verification. `check_L05.py` solves Cournot (30,30, profit 900), the joint maximum (45, 2025), the deviation $(\frac{135}{4},\,\frac{18225}{16})$, $\delta^*=\frac{81}{128}$ (both sides), the roots $\frac{70}{3}$ and $\frac{110}{3}$ at $\delta=\frac{1}{2}$, the profit $\frac{9100}{9}$, and $q_{\min}(\delta)$ with its values at $\frac{1}{2}$ and $\frac{81}{128}$.

(f) Common mistakes.

- Using the mixed stage-2 equilibrium (A with probability $\frac{3}{4}$, payoff 75) as the stick. The exercise asks for pure strategies; with the mixed stick the threshold would be $\frac{9}{16}=0.5625$, still above 0.5.
- Forgetting that the deviation against 22.5 is 33.75, not the Cournot quantity 30.

Answer. (a) Cournot $q_i = 30$, profit 900; stage 2 pure equilibria (A, a) and (B, b). (b) $q_i = 22.5$ (profit 1012.5) and (B, b). (c) $\delta \geq \frac{81}{128} \approx 0.633$. (d) At $\delta = 0.5$: $q_i = \frac{70}{3}$ (profit $\frac{9100}{9}$) in stage 1, then (B, b). (e) The lowest sustainable quantity is $q_{\min} = 30 - \frac{20\sqrt{2}}{3}\sqrt{\delta} \to 30$, the Cournot level, so profits fall to 900 in stage 1 while stage 2 stays at (B, b).

Item 20: MWG Exercise 12.AA.1

Advanced · difficulty 3/5

Statement (verbatim from the list, item 20)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Appendix A to Ch. 12 "Infinitely Repeated Games and the Folk Theorem", Exercise 12.AA.1ˆA (in text), p. 420 [PDF 221]. Answer in the source: in the MWG solutions manual (which marks the printed statement as incorrect: first-printing erratum). Reproduced verbatim. (Context, not in the exercise: the stage game is a duopoly in which firm i chooses q_i and earns $\pi_i(q_1, q_2)$; $q^* = (q_1^*, q_2^*)$ is its Nash equilibrium; Nash reversion, Definition 12.AA.1 on p. 418, plays a prescribed outcome path until someone deviates and q^* forever after.)

> **Exercise 12.AA.1:** Argue that no pair of actions q such that $\pi_i(q_1, q_2) < \pi_i(q_1^*, q_2^*)$ for some i can be sustained as a stationary SPNE outcome path using Nash reversion.

Note (not part of the exercise). The MWG solutions manual (p. 12-46) says the printed statement is incorrect (first-printing erratum), gives a counterexample, and states the closest correct claim: no q with $\pi_i(q) < \pi_i(q^*)$ for *all* i can be sustained by Nash reversion. The solution works both.

(a) What is asked. Argue that no action pair q with $\pi_i(q) < \pi_i(q^*)$ for some i can be a stationary SPNE outcome path using Nash reversion. *As printed the claim is false*; the MWG solutions manual (p. 12-46) marks it as a first-printing erratum. This solution shows why, and proves what is true.

(b) Method. One-stage deviation principle for Nash reversion with normalised payoffs (Lecture 5, slides 21, 24–25, 28–29; MWG Lemma 12.AA.1 and condition (12.AA.3), pp. 418–419; Tadelis §10.3, pp. 196–202). Notation: $q^* = (q_1^*, q_2^*)$ is the *unique* stage Nash equilibrium, $\tilde{\pi}_i(q_j) = \max_{q_i'} \pi_i(q_i', q_j)$ is player i's best-reply payoff against q_j.

(c) Solution. Step 1: the exact condition. Infinite repetition of q with reversion to q^* is an SPNE iff no player gains from a one-period deviation in the cooperative state (the punishment state is a stage equilibrium). A deviation is an action $q_i' \neq q_i$:

$$\pi_i(q) \;\geq\; (1-\delta) \max_{q_i' \neq q_i} \pi_i(q_i', q_j) + \delta\, \pi_i(q^*), \qquad i = 1, 2. \tag{2}$$

MWG's (12.AA.3) writes $\tilde{\pi}_i(q_j)$ instead of $\max_{q_i' \neq q_i}$. The two differ only when q_i is *already* a best reply to q_j: then $\tilde{\pi}_i(q_j) = \pi_i(q)$ is not a deviation at all, and (12.AA.3) reads $0 \leq \frac{\delta}{1-\delta}[\pi_i(q) - \pi_i(q^*)]$, which is stricter than the true condition (2). The claim in the exercise rests on the stricter form.

Step 2: a counterexample (the manual's). Stage game:

	L	R
U	$\underline{-1},\ 9$	$-1,\ \underline{10}$
D	$-10,\ -1$	$\underline{0},\ \underline{0}$

Cell by cell: at (U, L) player 2 gains by R ($10 > 9$); at (U, R) player 1 gains by D ($0 > -1$); at (D, L) player 1 gains by U ($-1 > -10$); (D, R) is an equilibrium. So $q^* = (D, R)$ is the unique equilibrium, with payoff $(0, 0)$. Take $q = (U, L)$ with $\pi_1(q) = -1 < 0 = \pi_1(q^*)$. Check (2):

- *Player 1* (U is already his best reply to L; the only deviation is D): $-1 \geq (1-\delta)(-10) + \delta \cdot 0 \iff \delta \leq \frac{9}{10}$.

- *Player 2* (deviation R, payoff 10): $9 \geq (1-\delta)\,10 + \delta \cdot 0 \iff \delta \geq \frac{1}{10}$.

So (U, L) forever is an SPNE outcome path with Nash reversion iff $\frac{1}{10} \leq \delta \leq \frac{9}{10}$, although player 1's payoff is below his Nash payoff. Player 1 stays only because deviating costs him 9 at once and the reward (payoff 0 in the future) is far off: he must be *impatient* enough.

Step 3: what is true. *(i)* If $\pi_i(q) < \pi_i(q^*)$ and q_i is *not* a best reply to q_j, then q cannot be sustained, for any δ: deviating to a best reply gives $\tilde{\pi}_i(q_j) > \pi_i(q)$ now and $\pi_i(q^*) > \pi_i(q)$ forever after, so the deviation payoff is strictly larger than $\pi_i(q) = (1-\delta)\pi_i(q) + \delta\pi_i(q)$.

(ii) The corrected statement (the manual's): no q with $\pi_i(q) < \pi_i(q^*)$ for *all* $i = 1, 2$ can be sustained. Proof: q^* is the unique equilibrium and $q \neq q^*$ (the payoffs differ), so q is not an equilibrium and some player j has q_j not a best reply to q_{-j}; since $\pi_j(q) < \pi_j(q^*)$, part (i) applies to j. Equivalently: if q_i were a best reply for both players, q would be a stage equilibrium.

(iii) The printed "some i" version holds when that player's action is not a best reply, as in (i).

Step 4: a Cournot illustration. Stage game $P = 100 - q_1 - q_2$, cost $10q_i$: $q^* = (30, 30)$, profit 900. Take $q = (40, 40)$: $\pi_i(q) = 400 < 900$ for both. The best reply to 40 is 25 with profit 625, and $400 \geq (1-\delta)625 + \delta 900$ is false for every $\delta \in (0, 1)$ (the right side exceeds 625).

(d) Interpretation. Nash reversion is a *reward* for a player whose path payoff is below Nash, so it can only deter a deviator who is already getting more now than on the path. The exercise is true when both players do worse than Nash, or when the unhappy player has a profitable deviation.

(e) Verification. `check_L05.py` finds that (D, R) is the only pure equilibrium of the counterexample, the two inequalities hold at $\delta = \frac{1}{10}, \frac{1}{2}, \frac{9}{10}$ and fail at 0.91 (player 1) and 0.09 (player 2). It also checks on 300 random 3×3 games with a unique pure equilibrium that no profile below the equilibrium payoff for both players is sustainable for $\delta = 0.01, \ldots, 0.99$, and the Cournot illustration.

Answer. The claim for "some i" is false (manual's erratum): with the game above, (U, L) gives player 1 the payoff $-1 < 0$ and is sustainable by Nash reversion iff $\frac{1}{10} \leq \delta \leq \frac{9}{10}$. True versions: a player below his Nash payoff whose action is not a best reply always deviates (the deviation pays more now and Nash pays more later); and no q with $\pi_i(q) < \pi_i(q^*)$ for *both* players is sustainable, because with a unique Nash equilibrium one of them is not best-replying.

Item 21: Tadelis Exercise 10.7

Advanced · difficulty 4/5

Statement (verbatim from the list, item 21)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.7 "Diluted Happiness", pp. 217–218 [PDF 236–237]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Diluted Happiness:** Consider a relationship between a bartender and a customer. The bartender serves bourbon to the customer and chooses $x \in [0,1]$, which is the proportion of bourbon in the drink served, while $1-x$ is the proportion of water. The cost of supplying such a drink (standard 4-ounce glass) is cx, where $c>0$. The customer, without knowing x, decides on whether or not to buy the drink at the market price p. If he buys the drink his payoff is $vx-p$, and the bartender's payoff is $p-cx$. Assume that $v>c$ and all payoffs are common knowledge. If the customer does not buy the drink he gets 0 and the bartender gets $-cx$. Because the customer has some experience, once the drink is bought and he tastes it, he learns the value of x, but this is only after he pays for the drink.
>
> a. Find all the Nash equilibria of this game.
> b. Now assume that the customer is visiting town for 10 days, and this "bar game" will be played on each of the 10 evenings that the customer is in town. Assume that each player tries to maximize the (nondiscounted) sum of his stage payoffs. Find all the subgame-perfect equilibria of this game.
> c. Now assume that the customer is a local, and the players perceive the game as repeated infinitely many times. Assume that each player tries to maximize the discounted sum of his stage payoffs, where the discount rate is $\delta \in (0,1)$. What is the range of prices p (expressed in the parameters of the problem) for which there exists a subgame-perfect equilibrium in which every day the bartender chooses $x=1$ and the customer buys at the price p?
> d. For which values of δ (expressed in the parameters of the problem) can the price range that you found in (c) exist?

(a) What is asked. In the bartender–customer game: (a) all stage Nash equilibria; (b) all SPE when it is played on 10 evenings without discounting; (c) with infinite repetition, the prices p at which an SPE has $x=1$ and a purchase every day; (d) the values of δ for which that price range is non-empty.

(b) Method. (a) Strict dominance (Tadelis §4.1). (b) Finite repetition of a game with a unique stage equilibrium: Prop. 10.1, p. 192 (§10.1). (c)–(d) Grim trigger and the one-stage deviation principle (Prop. 10.3, p. 198; (10.3) p. 200; (10.5) p. 202), applied to each player separately because the game is asymmetric, plus the minimax argument showing the punishment is the harshest available (`Notas/L05/L05-05-critical-discount-factor.md`, §1 and §5). Slides: Lecture 4, slides 21–22 (repeated games, $G(T,\delta)$) and slide 17 (SPE). Lecture 5 has no slides yet.

(c) Solution. Notation. The bartender (B) chooses $x \in [0,1]$; the customer (K) buys with probability $\beta \in [0,1]$, without seeing x, so within a day the moves are effectively simultaneous. Stage payoffs:

$$u_B(x,\beta) = \beta p - cx, \qquad u_K(x,\beta) = \beta(vx-p),$$

with $v > c > 0$. The price p is fixed by the market; take $p > 0$ (the case $p = 0$ is noted where it matters). The customer learns x only if he buys. $\delta \in (0,1)$ is the discount factor (the book's "discount rate").

(a) Stage Nash equilibria. For the bartender, $\partial u_B/\partial x = -c < 0$ whatever β is, so $x = 0$ is strictly dominant: a diluted drink is cheaper, and the customer cannot see it before paying. Against $x = 0$, buying gives the customer $v \cdot 0 - p = -p < 0$ and not buying gives 0, so he does not buy ($\beta = 0$). The unique Nash equilibrium is

$$(x, \beta) = (0, 0), \qquad \text{payoffs } (u_B, u_K) = (0, 0).$$

(If $p = 0$ the customer is indifferent, and every $\beta \in [0,1]$ with $x = 0$ is an equilibrium, all with payoffs $(0,0)$.) The efficient outcome, $x = 1$ with a purchase, creates surplus $v - c > 0$ but is not an equilibrium: this is a one-sided Prisoner's Dilemma (a trust game).

(b) Ten evenings. The stage game has a unique Nash equilibrium, so by backward induction (Prop. 10.1) the unique SPE is $x = 0$ and no purchase on every evening after every history. Step by step: on evening 10 nothing follows, $x = 0$ is strictly dominant and so the customer does not buy, whatever he knows about earlier evenings. On evening $k < 10$, by induction play from $k+1$ on is (0, no purchase) whatever happens on evening k, so evening k is the stage game plus a constant, and the same argument applies. The customer's imperfect information about x on days he does not buy does not matter, because $x = 0$ is strictly dominant at every information set. (With $p = 0$ the bartender's payoff on every evening is $-cx$ whatever the customer does, so he still chooses $x = 0$ every evening, and the customer may buy or not.) Payoffs: $(0,0)$.

(c) Infinite repetition: the price range. Strategies (grim trigger):

- bartender: $x = 1$ on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise $x = 0$ forever;
- customer: buy on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise never buy again.

On the path the customer buys every day, so he observes every x and any dilution is detected; a refusal to buy is seen by the bartender. In the punishment phase the customer no longer observes x, but nothing there depends on it.

Bartender, cooperation phase. Staying gives $p - c$ every day: $\frac{p-c}{1-\delta}$. The customer buys today without seeing x, so the best deviation is $x = 0$ (any $x \in (0,1)$ earns $p - cx < p$). It pays p today and 0 forever after:

$$\frac{p-c}{1-\delta} \;\geq\; p \iff p - c \geq (1-\delta)p \iff \delta p \geq c \iff p \;\geq\; \frac{c}{\delta}.$$

Customer, cooperation phase. Staying gives $v - p$ every day: $\frac{v-p}{1-\delta}$. The deviation, not buying, gives 0 today and 0 forever:

$$\frac{v-p}{1-\delta} \;\geq\; 0 \iff p \;\leq\; v.$$

Punishment phase. The prescribed play is the stage equilibrium (0, no purchase) and the future does not depend on today's action, so a deviation only changes today's payoff: the bartender would pay $cx > 0$ for nothing, and the customer would pay p for a drink with $x = 0$. Neither gains.

So the grim-trigger profile is an SPE with $x = 1$ and a purchase every day iff

$$\frac{c}{\delta} \;\leq\; p \;\leq\; v.$$

No other SPE does better. The bartender can guarantee himself 0 every day by choosing $x = 0$ (his minimax payoff, reached when the customer stops buying), and the customer can guarantee 0

by not buying. Grim trigger already pushes each deviator to 0, so no other punishment can widen the range: $[c/\delta, v]$ is the set of prices for which *any* SPE with this path exists.

(d) When the range is non-empty.

$$\frac{c}{\delta} \leq v \iff \delta \geq \frac{c}{v}.$$

Because $v > c$, $\frac{c}{v} < 1$, so the condition holds for all $\delta \in [\frac{c}{v}, 1)$. At $\delta = \frac{c}{v}$ the only price is $p = v$, which leaves the customer no surplus. For example, with $c = 1$, $v = 3$, $\delta = \frac{1}{2}$: $p \in [2, 3]$.

(d) Interpretation. Honest service is sustained by the value of the relationship, and that value has to come from the price. The bartender must earn a premium, $p - c \geq \frac{1-\delta}{\delta} c$ above cost, large enough that losing a regular customer outweighs the one-time saving c from watering one drink. A tourist, on a known finite visit, is cheated on every drink and so never buys; a local, valued as a long-term customer, is served full drinks at a price above cost. This is the logic of quality-assuring prices and relational contracts (Cabral §7.3, p. 174): with p too low there is no rent to lose, and with p above v the customer would rather not buy. The more patient the bartender, the smaller the premium needed.

(e) Verification. `check_C_L05.py` checks with sympy that u_B falls in x for every β (so $x = 0$ is strictly dominant) and that the bartender's condition reduces to $\delta p - c \geq 0$. For $(v, c) = (3, 1)$ and $(10, 9)$ and $\delta \in \{\frac{1}{2}, \frac{9}{10}, \frac{c}{v}, \frac{c}{v} + \frac{1}{1000}\}$ it confirms in exact fractions that both incentive constraints hold at $p = c/\delta$ and $p = v$ and fail at $c/\delta - \frac{1}{1000}$ and $v + \frac{1}{1000}$, or that no price works when $c/\delta > v$. On the grid $\delta = k/1000$ it checks that the range is non-empty exactly when $\delta \geq c/v$. Output: "Tadelis 10.7: stage NE x = 0, no purchase (p > 0); SPE price range c/delta <= p <= v; non-empty iff delta >= c/v (checked at c/v and +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Letting the customer react to x on the same day. He pays before he tastes, so the bartender's deviation earns the full price p that day; the payoff from deviating is p, not 0.
- Giving only the bartender's constraint ($p \geq c/\delta$) and forgetting the customer's ($p \leq v$), or the reverse; the game is asymmetric and each player has his own condition.
- In (b), arguing that ten evenings is "long enough" for reputation to work. With a known last evening and a unique stage equilibrium the game unravels completely.

Answer. (a) $x = 0$ is strictly dominant, so for $p > 0$ the unique NE is $x = 0$ and no purchase, payoffs $(0, 0)$ (for $p = 0$, $x = 0$ with any purchase probability). (b) Unique SPE: $x = 0$ and no purchase on all 10 evenings after every history. (c) With grim trigger, the bartender needs $\frac{p-c}{1-\delta} \geq p$, i.e. $p \geq c/\delta$, and the customer needs $p \leq v$; since these punishments are each player's minimax, the range is exactly $\frac{c}{\delta} \leq p \leq v$. (d) It is non-empty iff $\delta \geq c/v$ (which is below 1 because $v > c$).

Item 22: Tadelis Exercise 9.5

Advanced · difficulty 4/5

Statement (verbatim from the list, item 22)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 9.5 "Campaign Spending", pp. 188–189 [PDF 207–208]. Answer in the source: none (the book prints no solutions; the preface says about half are on Princeton UP's website, not checked). Reproduced verbatim.

> **Campaign Spending:** Two political candidates are destined to play the following two-stage game. Assume throughout that there is no discounting ($\delta = 1$). First, they compete in the primaries of their respective parties. Each candidate i can spend $s_i \geq 0$ resources on ads that reach out to voters, which in turn increases the probability that candidate i wins the race. Given a pair of spending choices (s_1, s_2), the probability that candidate i wins is given by $\frac{s_i}{s_1+s_2}$. If neither spends any resources then each wins with probability $\frac{1}{2}$. Each candidate values winning at a payoff of $16 > 0$, and the cost of spending s_i is equal to s_i. After each player observes the resources spent by the other and a winner in the primaries is selected, they can choose how to interact. Each can choose to be pleasant (P for player 1 and p for player 2) or nasty (N and n, respectively). At this stage both players prefer that they be nice to each other rather than nasty, but if one player is nasty then the other prefers to be nasty too. The payoffs from this stage are given by the following matrix (where $w > 0$):
>
> | Player 1 \ Player 2 | p | n |
> |---|---|---|
> | P | w, w | $-1, 0$ |
> | N | $0, -1$ | $0, 0$ |
>
> a. Find the unique Nash equilibrium of the first-stage game and the two pure-strategy Nash equilibria of the second-stage game.
>
> b. What are the Pareto-optimal outcomes of each stage-game?
>
> c. For which value of w can the players support the path of Pareto-optimal outcomes as a subgame-perfect equilibrium?
>
> d. Assume that $w = 1$. What is the "best" symmetric subgame-perfect equilibrium that the players can support?
>
> e. What happens to the best symmetric subgame-perfect equilibrium that the players can support as w changes? In what way is this related to the role played by a discount factor?

(a) What is asked. A contest (s_i spent, win probability $\frac{s_i}{s_1+s_2}$, prize 16) followed by a nice-or-nasty game with payoff w: find the stage equilibria and Pareto-optimal outcomes, the w that supports the Pareto path, the best symmetric SPE at $w = 1$, and how it changes with w.

(b) Method. Carrot and stick with $\delta = 1$ and a continuous first stage (Tadelis §9.4, pp. 180–184; Thm. 9.1, p. 185). The stage-2 carrot is (P, p) with payoff w and the stick is (N, n) with payoff 0, so w plays the role of δ [carrot − stick] (Lecture 5, slide 15). The deviation gain is computed from the best reply in the contest.

(c) Solution. (a) Stage equilibria. *Contest:* $v_i = \frac{16 s_i}{s_i+s_j} - s_i$. The first-order condition is $\frac{16 s_j}{(s_i+s_j)^2} = 1$, i.e. $s_i + s_j = 4\sqrt{s_j}$, so

$$s_i(s_j) = 4\sqrt{s_j} - s_j \qquad (s_j \leq 16).$$

In a symmetric equilibrium $s = 4\sqrt{s} - s \Rightarrow \sqrt{s} = 2$, so $s_1 = s_2 = 4$ with $v_i = 8 - 4 = 4$. This is the unique equilibrium: if $s_j = 0$, any $s_i > 0$ wins with probability one and there is no best reply (the supremum 16 is not attained), so $s_j = 0$ is not part of an equilibrium. *Stage 2:* (P, p) with (w, w) and (N, n) with $(0, 0)$ are equilibria: against p, P beats N since $w > 0$; against n, N beats P since $0 > -1$.

(b) Pareto-optimal outcomes. The prize is shared, so $v_1 + v_2 = 16 - s_1 - s_2$, maximal only at $s_1 = s_2 = 0$ (each wins with probability $\frac{1}{2}$): payoffs $(8, 8)$ against $(4, 4)$ at Nash. In stage 2, (P, p) with (w, w) Pareto-dominates (N, n).

(c) Supporting the Pareto path. Path: $s_1 = s_2 = 0$, then (P, p); after any deviation, (N, n). Against $s_j = 0$, any $s_i = \varepsilon > 0$ wins for sure and pays $16 - \varepsilon$, with supremum 16. Conforming gives $8 + w$, deviating gives at most $16 - \varepsilon$ plus the stick 0:

$$8 + w \;\geq\; 16 - \varepsilon \;\text{ for every } \varepsilon > 0 \iff w \;\geq\; 8.$$

(At $w = 8$ the deviation payoff $16 - \varepsilon$ is strictly below 16, so the path is an SPE.) The Pareto path is an SPE iff $w \geq 8$; the punishment state is a stage equilibrium and needs no check.

(d) $w = 1$**.** The path with (P, p) in stage 2 and $s_1 = s_2 = s$ in stage 1 has payoff $8 - s + w$. The best deviation against s is $x = 4\sqrt{s} - s$, with win probability $\frac{x}{x+s} = 1 - \frac{\sqrt{s}}{4}$ and payoff

$$16\Big(1 - \tfrac{\sqrt{s}}{4}\Big) - (4\sqrt{s} - s) = 16 - 8\sqrt{s} + s,$$

followed by (N, n) worth 0. The incentive constraint is

$$8 - s + w \geq 16 - 8\sqrt{s} + s \iff 2\big(\sqrt{s} - 2\big)^2 \leq w \iff 2 - \sqrt{\tfrac{w}{2}} \leq \sqrt{s} \leq 2 + \sqrt{\tfrac{w}{2}}.$$

The first-stage payoff $8 - s$ falls with s, so the best symmetric SPE uses the smallest admissible s:

$$s_{\min}(w) = \Big(2 - \sqrt{w/2}\Big)^2, \qquad s_{\min}(1) = \big(2 - \tfrac{1}{\sqrt{2}}\big)^2 = \tfrac{9}{2} - 2\sqrt{2} \approx 1.672,$$

and total payoff $8 - 1.672 + 1 \approx 7.328$ per player (against $4 + 1 = 5$ if they spend the Nash amount 4 and reach (P, p) anyway).

(e) As w **changes.** $s_{\min}(w)$ falls from 4 (the Nash level, as $w \to 0$) to 0 at $w = 8$ and stays at 0 for $w \geq 8$. A larger stage-2 gap supports less wasteful spending, and for $w \geq 8$ the whole Pareto path. The role of w is that of a discount factor: with discount factor δ the constraint reads $2(\sqrt{s} - 2)^2 \leq \delta w$, so what matters is δw, the discounted gap between carrot and stick, exactly as $\delta\,[\text{carrot} - \text{stick}]$ in the other items.

(d) Interpretation. The contest is a prisoner's dilemma in disguise: spending is wasted in the aggregate ($v_1 + v_2 = 16 - s_1 - s_2$). Stage 2 provides the incentive to restrain spending, and the amount that can be sustained moves smoothly with its size.

(e) Verification. `check_L05.py` verifies the best reply $4\sqrt{s_j} - s_j$, the Nash spending 4 with payoff 4, the deviation payoff $16 - 8\sqrt{s} + s$, the factorisation $2(\sqrt{s} - 2)^2 \leq w$, $s_{\min}(1) = \frac{9}{2} - 2\sqrt{2}$, $s_{\min}(8) = 0$, $s_{\min}(w) \to 4$ as $w \to 0$ and the stage-2 equilibria for $w \in \{1, 8, \frac{1}{2}\}$.

Answer. (a) Contest equilibrium $s_1 = s_2 = 4$, payoff 4; stage 2 equilibria (P, p) and (N, n). (b) Pareto-optimal: $s = (0, 0)$ (payoff 8 each) and (P, p). (c) $w \geq 8$. (d) At $w = 1$, $s_1 = s_2 = \frac{9}{2} - 2\sqrt{2} \approx 1.67$ with (P, p) in stage 2 (payoff ≈ 7.33). (e) $s_{\min}(w) = (2 - \sqrt{w/2})^2$ decreases from 4 to 0 as w goes from 0 to 8; w plays the role of the discounted gap $\delta\,[\text{carrot} - \text{stick}]$.

Item 23: Gibbons Problem 2.8

Advanced · difficulty 4/5

Statement (verbatim from the list, item 23)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems (Section 2.2), Problem 2.8, p. 133 [PDF 140]. Answer in the source: none in the book. Reproduced verbatim.

> 2.8. Modify the tournament model in Section 2.2.D so that worker i's output is $y_i = e_i - (1/2)s_j + \varepsilon_i$, where $s_j \geq 0$ represents sabotage by worker j, and worker i's disutility of (productive and destructive) effort is $g(e_i) + g(s_i)$, as in Lazear (1989). Show that the optimal prize $w_H - w_L$ is smaller than when there is no possibility of sabotage (as in the text).

Referred to in the statement. The tournament model of Section 2.2.D and its optimal prize (Gibbons pp. 79–82 [PDF p. 86–89]) (reproduced from the book):

> Consider two workers and their boss. Worker i (where $i = 1$ or 2) produces output $y_i = e_i + \varepsilon_i$, where e_i is effort and ε_i is noise. Production proceeds as follows. First, the workers simultaneously choose nonnegative effort levels: $e_i \geq 0$. Second, the noise terms ε_1 and ε_2 are independently drawn from a density $f(\varepsilon)$ with zero mean. Third, the workers' outputs are observed but their effort choices are not. The workers' wages therefore can depend on their outputs but not (directly) on their efforts.
>
> Suppose the workers' boss decides to induce effort from the workers by having them compete in a tournament, as first analyzed by Lazear and Rosen (1981). The wage earned by the winner of the tournament (i.e., the worker with the higher output) is w_H; the wage earned by the loser is w_L. The payoff to a worker from earning wage w and expending effort e is $u(w, e) = w - g(e)$, where the disutility of effort, $g(e)$, is increasing and convex (i.e., $g'(e) > 0$ and $g''(e) > 0$). The payoff to the boss is $y_1 + y_2 - w_H - w_L$.
>
> [...] Suppose the boss has chosen the wages w_H and w_L. If the effort pair (e_1^*, e_2^*) is to be a Nash equilibrium of the remaining game between the workers then, for each i, e_i^* must maximize worker i's expected wage, net of the disutility of effort: e_i^* must solve

$$\begin{aligned}\max_{e_i \geq 0} \; & w_H \operatorname{Prob}\{y_i(e_i) > y_j(e_j^*)\} + w_L \operatorname{Prob}\{y_i(e_i) \leq y_j(e_j^*)\} - g(e_i) \\ & = (w_H - w_L) \operatorname{Prob}\{y_i(e_i) > y_j(e_j^*)\} + w_L - g(e_i),\end{aligned} \tag{2.2.4}$$

> where $y_i(e_i) = e_i + \varepsilon_i$. The first-order condition for (2.2.4) is

$$(w_H - w_L) \frac{\partial \operatorname{Prob}\{y_i(e_i) > y_j(e_j^*)\}}{\partial e_i} = g'(e_i). \tag{2.2.5}$$

> [...] By Bayes' rule,

$$\begin{aligned}\operatorname{Prob}\{y_i(e_i) > y_j(e_j^*)\} &= \operatorname{Prob}\{\varepsilon_i > e_j^* + \varepsilon_j - e_i\} \\ &= \int_{\varepsilon_j} \operatorname{Prob}\{\varepsilon_i > e_j^* + \varepsilon_j - e_i \mid \varepsilon_j\} f(\varepsilon_j)\, d\varepsilon_j \\ &= \int_{\varepsilon_j} [1 - F(e_j^* - e_i + \varepsilon_j)] f(\varepsilon_j)\, d\varepsilon_j,\end{aligned}$$

> so the first-order condition (2.2.5) becomes

$$(w_H - w_L) \int_{\varepsilon_j} f(e_j^* - e_i + \varepsilon_j) f(\varepsilon_j)\, d\varepsilon_j = g'(e_i).$$

> In a symmetric Nash equilibrium (i.e., $e_1^* = e_2^* = e^*$), we have

$$(w_H - w_L) \int_{\varepsilon_j} f(\varepsilon_j)^2\, d\varepsilon_j = g'(e^*). \tag{2.2.6}$$

[...] We now work backwards to the first stage of the game. Suppose that if the workers agree to participate in the tournament (rather than accept alternative employment) then they will respond to the wages w_H and w_L by playing the symmetric Nash equilibrium characterized by (2.2.6). (We thus ignore the possibilities of asymmetric equilibria and of an equilibrium in which the workers' effort choices are given by the corner solution $e_1 = e_2 = 0$, rather than by the first-order condition (2.2.5).) Suppose also that the workers' alternative employment opportunity would provide utility U_a. Since in the symmetric Nash equilibrium each worker wins the tournament with probability one-half (i.e., $\text{Prob}\{y_i(e^*) > y_j(e^*)\} = 1/2$), if the boss intends to induce the workers to participate in the tournament then she must choose wages that satisfy

$$\tfrac{1}{2}w_H + \tfrac{1}{2}w_L - g(e^*) \geq U_a. \tag{2.2.7}$$

Assuming that U_a is low enough that the boss wants to induce the workers to participate in the tournament, she therefore chooses wages to maximize expected profit, $2e^* - w_H - w_L$, subject to (2.2.7). At the optimum, (2.2.7) holds with equality:

$$w_L = 2U_a + 2g(e^*) - w_H. \tag{2.2.8}$$

Expected profit then becomes $2e^* - 2U_a - 2g(e^*)$, so the boss wishes to choose wages such that the induced effort, e^*, maximizes $e^* - g(e^*)$. The optimal induced effort therefore satisfies the first-order condition $g'(e^*) = 1$. Substituting this into (2.2.6) implies that the optimal prize, $w_H - w_L$, solves

$$(w_H - w_L)\int_{\varepsilon_j} f(\varepsilon_j)^2\, d\varepsilon_j = 1,$$

and (2.2.8) then determines w_H and w_L themselves.

(a) What is asked. In the tournament of §2.2.D, let output be $y_i = e_i - \frac{1}{2}s_j + \varepsilon_i$ and effort cost $g(e_i) + g(s_i)$, where $s_j \geq 0$ is sabotage by the rival. Show that the optimal prize $w_H - w_L$ is smaller than without sabotage.

(b) Method. Backward induction in a two-stage game (Gibbons §2.2.A and §2.2.D, pp. 71–73, 79–82): solve the workers' simultaneous second-stage game for given wages, then the boss's wage choice (Lecture 5, slide 14: last stage first). Notation as in the text: ε_i are i.i.d. with an atomless density f and mean zero; $g' > 0$, $g'' > 0$; the payoff to a worker is $w - g(e_i) - g(s_i)$, the boss's is $y_1 + y_2 - w_H - w_L$; U_a is the alternative utility; we look at the symmetric interior equilibrium, as the text does.

(c) Solution. Step 1: the workers. Write $\Delta = w_H - w_L$. Worker i's output exceeds j's iff

$$y_i - y_j = (e_i - e_j) + \tfrac{1}{2}(s_i - s_j) + (\varepsilon_i - \varepsilon_j) > 0,$$

so $\Pr\{y_i > y_j\} = H(e_i - e_j + \frac{1}{2}(s_i - s_j))$, with H the c.d.f. of $\varepsilon_j - \varepsilon_i$ and density $h(z) = \int f(\varepsilon) f(\varepsilon + z)\, d\varepsilon$, $h(0) = \int f^2$. Worker i maximises $w_L + \Delta \Pr\{y_i > y_j\} - g(e_i) - g(s_i)$, with first-order conditions

$$\Delta\, h(\cdot) = g'(e_i), \qquad \tfrac{1}{2}\,\Delta\, h(\cdot) = g'(s_i).$$

(Own effort raises the probability of winning at rate h; own sabotage raises it at half that rate.) In the symmetric equilibrium $h = h(0)$ and

$$g'(e^*) = \Delta\, h(0), \qquad g'(s^*) = \tfrac{1}{2}\,\Delta\, h(0) = \tfrac{1}{2}\, g'(e^*).$$

So sabotage is positive (assume $g'(0)$ small enough) but smaller than effort, $s^* < e^*$, and both rise with the prize. Write $s^* = s(e^*)$, where $s(e)$ solves $g'(s) = \frac{1}{2}g'(e)$; $s'(e) = \dfrac{g''(e)}{2g''(s)} > 0$.

Step 2: the boss. Each worker wins with probability $\frac{1}{2}$, so participation requires $\frac{1}{2}w_H + \frac{1}{2}w_L - g(e^*) - g(s^*) \geq U_a$, binding at the optimum: $w_H + w_L = 2U_a + 2g(e^*) + 2g(s^*)$. Expected output is $2e^* - s^*$ (sabotage lowers it), so the profit is

$$\Pi(e) = 2e - s(e) - 2g(e) - 2g\big(s(e)\big) - 2U_a,$$

chosen through Δ, which fixes e via $g'(e) = \Delta h(0)$. *Without sabotage* the profit is $\Pi_0(e) = 2e - 2g(e) - 2U_a$, maximal at $g'(e_0) = 1$, and the optimal prize solves $\Delta_0\, h(0) = 1$.

Step 3: comparison. Differentiating, and using $2g'(s) = g'(e)$,

$$\Pi'(e) = 2 - s'(e) - 2g'(e) - 2g'(s)\, s'(e) = 2\big[1 - g'(e)\big] - s'(e)\big[1 + g'(e)\big].$$

For every e with $g'(e) \geq 1$ both terms are ≤ 0 and the second is strictly negative, so $\Pi'(e) < 0$; in particular at e_0, where $g'(e_0) = 1$, $\Pi'(e_0) = -2s'(e_0) < 0$. Hence any optimal e^* satisfies $g'(e^*) < 1$, i.e. $e^* < e_0$ (g' increasing), and the optimal prize is

$$\Delta^* = \frac{g'(e^*)}{h(0)} \;<\; \frac{1}{h(0)} = \Delta_0.$$

The first-order condition at the optimum confirms it: $2[1 - g'(e^*)] = s'(e^*)[1 + g'(e^*)] > 0$.

Example. $g(x) = \frac{x^2}{2}$: $s(e) = \frac{e}{2}$, $\Pi(e) = \frac{3}{2}e - \frac{5}{4}e^2$ (up to the constant), optimum $e^* = \frac{3}{5}$ against $e_0 = 1$: the optimal prize with sabotage is $\frac{3}{5}$ of the prize without.

(d) Interpretation. Sabotage is a pure waste: it costs effort $g(s)$ and lowers output by $\frac{1}{2}s$. A larger prize raises both useful effort and sabotage, so the boss tunes the prize down to curb sabotage, even though this also reduces productive effort.

(e) Verification. `check_L05.py` verifies $\Pi'(e) = 2(1 - g') - s'(1 + g')$ for $g(x) = x^2/2$, the optimum $e^* = \frac{3}{5}$ with $g'(e^*) = \frac{3}{5} < 1$, that $\Pi' < 0$ for $e \in [1, 4)$ in that example, and, numerically for $g(x) = \frac{x^2}{2} + \frac{x^4}{4}$ (solving $g'(s) = \frac{1}{2}g'(e)$), that the optimum has $g'(e^*) = 0.504 < 1 = g'(e_0)$.

Answer. In the symmetric equilibrium $g'(e^*) = \Delta h(0)$ and $g'(s^*) = \frac{1}{2}g'(e^*)$. The boss's profit $\Pi(e) = 2e - s(e) - 2g(e) - 2g(s(e))$ has $\Pi'(e) = 2[1 - g'(e)] - s'(e)[1 + g'(e)] < 0$ whenever $g'(e) \geq 1$, so the optimal induced effort has $g'(e^*) < 1$, whereas without sabotage $g'(e_0) = 1$. Since $\Delta = g'(e)/h(0)$, the optimal prize with sabotage is smaller than the one without it.

Item 24: Gibbons Problem 2.16

Advanced · difficulty 4/5

Statement (verbatim from the list, item 24)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §2.3 Problems, Problem 2.16, p. 136 [PDF 143]. Answer in the source: none in the book. Reproduced verbatim. (The one-shot model it builds on is Gibbons §2.1.C, pp. 64–68: the union sets the wage, then the firm sets employment.)

> 2.16. In the model of wages and employment analyzed in Section 2.1.C, the backwards-induction outcome is not socially efficient. In practice, however, a firm and a union negotiate today over the terms of a three-year contract, then negotiate three years from today over the terms of a second contract, and so on. Thus, the relationship may be more accurately characterized as a repeated game, as in Espinosa and Rhee (1989).
>
> This problem derives conditions under which a subgame-perfect Nash equilibrium in the infinitely repeated game is Pareto-superior to the backwards-induction outcome of the one-shot game. Denote the union's utility and the firm's profit in the backwards-induction outcome of the one-shot game by U^* and π^*, respectively. Consider an alternative utility-profit pair (U, π) associated with an alternative wage-employment pair (w, L). Suppose that the parties share the discount factor δ (per three-year period). Derive conditions on (w, L) such that (1) (U, π) Pareto-dominates (U^*, π^*) and (2) (U, π) is the outcome of a subgame-perfect Nash equilibrium of the infinitely repeated game, where (U^*, π^*) is played forever following a deviation.

Referred to in the statement. The model of wages and employment of Section 2.1.C (Gibbons pp. 64–66 [PDF p. 71–73]) (reproduced from the book):

> In Leontief's (1946) model of the relationship between a firm and a monopoly union (i.e., a union that is the monopoly seller of labor to the firm), the union has exclusive control over wages, but the firm has exclusive control over employment. (Similar qualitative conclusions emerge in a more realistic model in which the firm and the union bargain over wages but the firm retains exclusive control over employment.) The union's utility function is $U(w, L)$, where w is the wage the union demands from the firm and L is employment. Assume that $U(w, L)$ increases in both w and L. The firm's profit function is $\pi(w, L) = R(L) - wL$, where $R(L)$ is the revenue the firm can earn if it employs L workers (and makes the associated production and product-market decisions optimally). Assume that $R(L)$ is increasing and concave.
>
> Suppose the timing of the game is: (1) the union makes a wage demand, w; (2) the firm observes (and accepts) w and then chooses employment, L; (3) payoffs are $U(w, L)$ and $\pi(w, L)$. [. . .]
>
> First, we can characterize the firm's best response in stage (2), $L^*(w)$, to an arbitrary wage demand by the union in stage (1), w. Given w, the firm chooses $L^*(w)$ to solve
>
> $$\max_{L \geq 0} \pi(w, L) = \max_{L \geq 0} R(L) - wL,$$
>
> the first-order condition for which is
>
> $$R'(L) - w = 0.$$
>
> [. . .] To guarantee that the first-order condition $R'(L) - w = 0$ has a solution, assume that $R'(0) = \infty$ and that $R'(\infty) = 0$ [. . .]
>
> We turn next to the union's problem at stage (1). Since the union can solve the firm's second-stage problem as well as the firm can solve it, the union should anticipate that the firm's reaction to the wage demand w will be to choose the employment level $L^*(w)$. Thus, the union's problem at the first stage amounts to
>
> $$\max_{w \geq 0} U(w, L^*(w)).$$

(a) What is asked. In the infinitely repeated wage-employment game, derive conditions on (w, L) such that (1) (U, π) Pareto-dominates the backwards-induction outcome (U^*, π^*) and (2) (U, π) is the outcome of an SPNE in which (U^*, π^*) is played forever after a deviation.

(b) Method. The one-shot model is Gibbons §2.1.C (pp. 64–68): the union chooses w, the firm observes it and chooses L; the firm's profit is $\pi(w, L) = R(L) - wL$ and the union's utility is $U(w, L)$. The repeated game is the trigger strategy with the one-stage deviation principle and normalised payoffs (Lecture 5, slides 21, 24–29; Tadelis §10.3, pp. 196–202; Gibbons §2.3.B, pp. 88–91). Notation: $L(w)$ is the firm's best reply to w and (w^*, L^*) the backwards-induction outcome, $L^* = L(w^*)$, $w^* = \arg\max_w U(w, L(w))$.

(c) Solution. Strategies. Each period the union proposes w and the firm answers L; play (w, L) as long as it was played in every earlier period; after any other history play the one-shot backwards-induction behaviour (the union asks w^*, the firm plays $L(\cdot)$) forever. The punishment is an SPNE of the stage game, so no check is needed there. There are two cooperative-state checks, one per mover.

The union. If the union asks $w' \neq w$, the firm's strategy after this history is the punishment strategy: it best-replies, $L(w')$, in that period, and the backwards-induction outcome follows. So the union's deviation payoff is $(1-\delta)\,U(w', L(w')) + \delta U^*$, at most $(1-\delta)U^* + \delta U^* = U^*$, since U^* is the maximum of $U(w, L(w))$. The union conforms iff

$$U(w, L) \;\geq\; U^*.$$

The firm. After the union has asked w, the firm's best deviation is $L(w)$, with profit $\pi^d(w) = \max_{L'} [R(L') - wL'] \geq \pi(w, L)$. It conforms iff

$$\pi(w, L) \;\geq\; (1-\delta)\,\pi^d(w) + \delta\,\pi^* \iff \delta \;\geq\; \frac{\pi^d(w) - \pi(w, L)}{\pi^d(w) - \pi^*}.$$

(If $L = L(w)$ the numerator is zero and there is nothing to deter.) The firm's condition implies $\pi(w, L) \geq \pi^*$: if $\pi^d(w) \geq \pi^*$, the right side is at least π^*; if $\pi^d(w) < \pi^*$, the right side exceeds $\pi^d(w) \geq \pi$ and the condition fails.

Conditions on (w, L). Both properties together:

(i) $U(w, L) \geq U^*$ and $\pi(w, L) \geq \pi^*$, with at least one strict (Pareto dominance, property 1);

(ii) $\delta \geq \dfrac{\pi^d(w) - \pi(w, L)}{\pi^d(w) - \pi^*}$, where $\pi^d(w) = \max_{L'}[R(L') - wL']$ (property 2).

The union's constraint is $U \geq U^*$, already in (i); only the firm needs patience, because the union moves first and is automatically best-replied by the punishment.

Example (the model of item 3, M-G Ch. 4 Ex. 18). $U = wL$, $R(L) = L - \frac{L^2}{2}$ (the union's wage $w \in [0, 1]$ and hiring $L \in [0, 1]$): $L(w) = 1 - w$, $(w^*, L^*) = (\frac{1}{2}, \frac{1}{2})$, $U^* = \frac{1}{4}$, $\pi^* = \frac{1}{8}$. Total surplus $L - \frac{L^2}{2}$ is largest at $L = 1$, where $U = w$ and $\pi = \frac{1}{2} - w$. Pareto dominance requires $w \in [\frac{1}{4}, \frac{3}{8}]$. The firm's best deviation profit is $\pi^d(w) = \frac{(1-w)^2}{2}$, so

$$\delta(w) = \frac{\frac{(1-w)^2}{2} - \left(\frac{1}{2} - w\right)}{\frac{(1-w)^2}{2} - \frac{1}{8}} = \frac{w^2/2}{\frac{(1-w)^2}{2} - \frac{1}{8}}.$$

At $w = \frac{1}{4}$ (the union exactly as well off as at w^*, the firm much better off): $\delta = \frac{1}{5}$. At $w = \frac{3}{10}$: $\delta = \frac{3}{8}$. The threshold rises with w and reaches 1 at $w = \frac{3}{8}$, where the firm's gain from cooperation, $\pi - \pi^*$, vanishes.

(d) Interpretation. The one-shot outcome is inefficient because the union cannot commit to a lower wage with $L = 1$. Repetition lets the firm trust the union's wage by threatening to return to the one-shot outcome. The firm is the one who must be patient: its temptation is to hire fewer workers than promised ($L = 1$ at the wage w is not its best reply).

(e) Verification. `check_L05.py` verifies, in the example, $L(w) = 1 - w$, $\pi^d(w) = \frac{(1-w)^2}{2}$, the Pareto range $[\frac{1}{4}, \frac{3}{8}]$, $\delta(\frac{1}{4}) = \frac{1}{5}$, $\delta(\frac{3}{10}) = \frac{3}{8}$, $\delta(\frac{3}{8}) = 1$, monotonicity of $\delta(w)$, that the firm's constraint holds at $\delta = \frac{1}{5}$ and fails at 0.199 for $w = \frac{1}{4}$, and that the union's best deviation payoff is $U^* = \frac{1}{4}$ (grid).

Answer. (i) $U(w, L) \geq U^*$ and $\pi(w, L) \geq \pi^*$, one strictly; (ii) $\delta \geq \dfrac{\pi^d(w) - \pi(w, L)}{\pi^d(w) - \pi^*}$ with $\pi^d(w) = \max_{L'}[R(L') - wL']$ the firm's best-deviation profit. The union's deviation yields at most U^*, so its constraint is $U \geq U^*$. Example ($U = wL$, $R = L - L^2/2$): $(w, L) = (\frac{1}{4}, 1)$ is sustained iff $\delta \geq \frac{1}{5}$.

Item 25: MWG Exercise 12.AA.2

Advanced · difficulty 5/5

Statement (verbatim from the list, item 25)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 12 Exercises, Exercise 12.AA.2ˆC, p. 435 [PDF 228]. Answer in the source: in the MWG solutions manual. Reproduced verbatim. **Added (not in the exercise):** the statement of Proposition 12.AA.4 and the definition of the minimax payoff it uses, both reproduced verbatim from p. 421, so the item can be worked without the book.

> **12.AA.2ˆC** Prove Proposition 12.AA.4. [*Hint:* Consider a strategy profile of the following form: the players are to play an outcome path involving some pair (q_1, q_2) in period 1 and (q_1^*, q_2^*) in every period thereafter. If either player deviates, this outcome path is restarted.]

Referred to in the statement. The model of Appendix A, "Infinitely Repeated Games and the Folk Theorem" (MWG pp. 417–418 [PDF p. 219–220]), and the minimax payoff and Proposition 12.AA.4 (p. 421 [PDF p. 221]) (reproduced from the book; footnote 31 dropped):

> An infinitely repeated game consists of an infinite sequence of repetitions of a one-period simultaneous-move game, known as the *stage game*. For expositional simplicity, we focus here on the case in which there are two players.
>
> In the one-period stage game, each player i has a compact strategy set S_i; $q_i \in S_i$ is a particular feasible action for player i. Denote $q = (q_1, q_2)$ and $S = S_1 \times S_2$. Player i's payoff function is $\pi_i(q_i, q_j)$. We restrict our attention throughout to pure strategies. It will be convenient to define player i's one-period best-response payoff given that his rival plays q_j by $\hat{\pi}_i(q_j) = \text{Max}_{q \in S_i}\, \pi_i(q, q_j)$. We assume that the stage game has a unique pure strategy Nash equilibrium $q^* = (q_1^*, q_2^*)$ (the assumption of uniqueness is for expositional simplicity only).
>
> In the infinitely repeated game, actions are taken and payoffs are earned at the beginning of each period. The players discount payoffs with discount factor $\delta < 1$. Players observe each other's action choices in each period and have perfect recall. A pure strategy in this game for player i, s_i, is a sequence of functions $\{s_{it}(\cdot)\}_{t=1}^{\infty}$ mapping from the history of previous action choices (denoted H_{t-1}) to his action choice in period t, $s_{it}(H_{t-1}) \in S_i$. The set of all such pure strategies for player i is denoted by Σ_i, and $s = (s_1, s_2) \in \Sigma_1 \times \Sigma_2$ is a profile of pure strategies for the two players.
>
> Any pure strategy profile $s = (s_1, s_2)$ induces an *outcome path* $Q(s)$, an infinite sequence of actions $\{q_t = (q_{1t}, q_{2t})\}_{t=1}^{\infty}$ that will actually be played when the players follow strategies s_1 and s_2. Player i's discounted payoff from outcome path Q is given by $v_i(Q) = \sum_{t=0}^{\infty} \delta^t \pi_i(q_{1+t})$. We also define player i's *average payoff* from outcome path Q to be $(1-\delta)v_i(Q)$: this is the per-period payoff that, if infinitely repeated, would give player i a discounted payoff of $v_i(Q)$. Finally, it is also useful to define the discounted continuation payoff from outcome path Q from some period t onward (discounted to period t) by $v_i(Q, t) = \sum_{\tau=0}^{\infty} \delta^\tau \pi_i(q_{t+\tau})$.
>
> [...] To consider this issue, it is useful to let $\underline{\pi}_i = \text{Min}_{q_j}\, [\text{Max}_{q_i}\, \pi_i(q_i, q_j)]$ denote player i's *minimax payoff.*
>
> **Proposition 12.AA.4:** Consider an infinitely repeated game with $\delta > 0$ and $S_i \subset \mathbb{R}$ for $i = 1, 2$. Suppose also that $\pi_i(q)$ is differentiable at $q^* = (q_1^*, q_2^*)$, with $\partial \pi_i(q_i^*, q_j^*)/\partial q_j \neq 0$ for $j \neq i$ and $i = 1, 2$, and that $\pi_i(q_1^*, q_2^*) > \underline{\pi}_i$ for $i = 1, 2$. Then there is some SPNE with discounted payoffs to the two players of (v_1', v_2') such that $(1-\delta)v_i' < \pi_i(q_1^*, q_2^*)$ for $i = 1, 2$.

Optional extras.

- Gibbons 2.12 (p. 135): strategy, subgame and SPE in a repeated game; it is item 1 of the Lecture 4 list.

- Muñoz-Garcia Ch. 6 Ex. 7 (p. 299): the game of item 16 with a full worked answer — check item 16 against it.

- Tadelis 10.12 (p. 219): the folk theorem on the trust game, after Lecture 6 introduces that game.

- Cabral 7.15 (p. 182): grim strategies on an advertising agreement, as a bridge to Lecture 6's collusion.

(a) What is asked. Prove Proposition 12.AA.4: under the stated conditions there is an SPNE with discounted payoffs v_i' such that $(1-\delta)v_i' < \pi_i(q^*)$ for $i = 1, 2$, that is, punishments harsher than Nash reversion.

(b) Method. The hint's profile ("play q once, then q^* forever, and restart after a deviation") checked with the one-stage deviation principle and normalised payoffs (Lecture 5, slides 21, 24–25; MWG App. A to Ch. 12, pp. 418–421; solution manual, Ex. 12.AA.2). Notation: $\tilde{\pi}_i(q_j) = \max_{q_i'} \pi_i(q_i', q_j)$. As in the text, q^* is the stage Nash equilibrium, interior, so $\partial\pi_i(q^*)/\partial q_i = 0$; write $a_i = \partial\pi_i(q^*)/\partial q_j \neq 0$ $(j \neq i)$. The minimax assumption is not used in the construction: the incentive constraints below force $(1-\delta)v_i' \geq \tilde{\pi}_i(q_j) \geq \underline{\pi}_i$ by themselves.

(c) Solution. Step 1: two first-order facts. By the envelope theorem $\tilde{\pi}_i'(q_j^*) = a_i$. For a small move $q = q^* + \varepsilon d$ $(d = (d_1, d_2))$, the first-order condition $\partial\pi_i/\partial q_i = 0$ at q^* gives

$$\pi_i(q) - \pi_i(q^*) = a_i\, \varepsilon d_j + o(\varepsilon), \qquad \tilde{\pi}_i(q_j) - \pi_i(q^*) = a_i\, \varepsilon d_j + o(\varepsilon),$$

the same first-order term, because own deviations from q_i^* have no first-order effect.

Step 2: the direction. Choose $d_j = -\operatorname{sign}(a_i)$ for $i \neq j$ (player j's move hurts i). Then $x_i \equiv a_i \varepsilon d_j = -|a_i|\varepsilon < 0$, and for ε small $\pi_i(q) < \pi_i(q^*)$ for both i. Fix such a q.

Step 3: the profile. Two states. *Start:* play q. *Nash:* play q^*. From Start, if both played q, go to Nash; from Nash, if both played q^*, stay; after any other profile, in either state, go back to Start (restart). Writing $V_i = (1-\delta)v_i'$ for the normalised value of Start,

$$V_i = (1-\delta)\pi_i(q) + \delta\pi_i(q^*) \;<\; \pi_i(q^*) \qquad (\delta > 0),$$

which is the claim, if the profile is an SPNE.

Step 4: Nash state. Staying gives $\pi_i(q^*)$. A deviation yields at most $\pi_i(q^*)$ now (q^* is a stage equilibrium) and then Start, worth $V_i < \pi_i(q^*)$; the deviation payoff is below $\pi_i(q^*)$. No gain.

Step 5: Start state. Staying gives V_i. The best deviation gives $\tilde{\pi}_i(q_j)$ now and Start again: $(1-\delta)\tilde{\pi}_i(q_j) + \delta V_i$ (if q_i is already a best reply, every deviation gives less). Staying is optimal iff

$$V_i \;\geq\; \tilde{\pi}_i(q_j) \iff \delta\big[\pi_i(q^*) - \pi_i(q)\big] \;\geq\; \tilde{\pi}_i(q_j) - \pi_i(q).$$

By Step 1 the right side is $o(\varepsilon)$ (both $\tilde{\pi}_i(q_j)$ and $\pi_i(q)$ differ from $\pi_i(q^*)$ by $x_i + o(\varepsilon)$), while the left side is $\delta\,(|a_i|\varepsilon + o(\varepsilon))$, first order in ε. So for every $\delta > 0$ there is an $\bar{\varepsilon}(\delta) > 0$ such that for $\varepsilon \leq \bar{\varepsilon}$ the inequality holds for $i = 1, 2$.

Conclusion. For small enough ε the profile is an SPNE (Steps 4–5, one-stage deviation principle) with $(1-\delta)v_i' = V_i < \pi_i(q^*)$ for $i = 1, 2$. □

Cournot illustration. $P = 100 - q_1 - q_2$, cost $10q_i$: $q^* = (30, 30)$, $\pi_i^* = 900$, $a_i = -30 < 0$, so $d_j = +1$: both firms over-produce. With $q = (30+\varepsilon, 30+\varepsilon)$: $\pi_i(q) = 900 - 30\varepsilon - 2\varepsilon^2$, $\tilde{\pi}_i(q_j) = 900 - 30\varepsilon + \frac{\varepsilon^2}{4}$ (best reply $\frac{60-\varepsilon}{2}$), and the Start constraint becomes $\varepsilon\big[30\delta - \varepsilon\big(\frac{9}{4} - 2\delta\big)\big] \geq 0$, i.e. $\varepsilon \leq \frac{30\delta}{9/4 - 2\delta}$. For $\delta = \frac{1}{2}$: $\varepsilon \leq 12$. Take $q = (40, 40)$: $\pi_i(q) = 400$, $\tilde{\pi}_i = 625$, $V_i = \frac{1}{2}\cdot 400 + \frac{1}{2}\cdot 900 = 650 \geq 625$, and $V_i = 650 < 900$: the average payoff in the SPNE is below the Nash payoff, which Nash reversion alone cannot achieve (Exercise 12.AA.1, item 20). At the boundary $q = (42, 42)$: $\pi_i = 252$, $\tilde{\pi}_i = 576 = V_i$.

(d) Interpretation. Nash reversion punishes a deviator only by taking away the gains of cooperation. Restarting the path punishes by repeating the *costly* first period, which lowers both players' payoffs below Nash; because the costly period is one in which each player is playing (almost) a best reply, it is credible. Deviating from it gains only second-order amounts.

(e) Verification. `check_L05.py` verifies for the Cournot example $\pi_i(q) = 900 - 30\varepsilon - 2\varepsilon^2$, $\tilde{\pi}_i = 900 - 30\varepsilon + \varepsilon^2/4$, the factorisation of $V_i - \tilde{\pi}_i$, and for $\delta \in \{0.01, 0.1, 0.5, 0.9\}$ with ε half the bound: $\pi_i(q) < 900$, Start constraint, Nash-state constraint and $V_i < 900$ with exact fractions.

Answer. Proof: take $q = q^* + \varepsilon d$ with $d_j = -\operatorname{sign}(\partial \pi_i / \partial q_j)$, so $\pi_i(q) < \pi_i(q^*)$ for both players. Play q once, then q^* forever, restarting after any deviation. In the Nash state a deviation gains nothing now and restarts a path worth less than $\pi_i(q^*)$; in the Start state the deviation gain $\tilde{\pi}_i(q_j) - \pi_i(q)$ is $o(\varepsilon)$ while the loss $\delta[\pi_i(q^*) - \pi_i(q)]$ is first order. So the profile is an SPNE with $(1-\delta)v_i' = (1-\delta)\pi_i(q) + \delta\pi_i(q^*) < \pi_i(q^*)$.